



Bahir Dar University

Bahir Dar Institute of Technology

Faculty of Computing

Requirement Analysis Document (RAD) for Industrial project on

Therapy for treating phobias and Anxiety

Submitted to the faculty of computing in partial fulfillment of the requirements for the degree of Bachelor of Science in Computer Science

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2026

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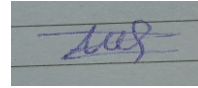
Declaration

We here by declare that this project is our original work and has not been submitted for a degree or diploma in any other university or institution. All sources of information and materials used in this project have been properly cited and acknowledged.

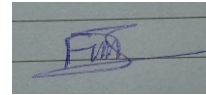
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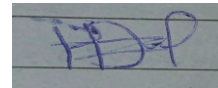
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Sequence and Activity modeling		√	√
UI/UX Design	√	√	
Persistent and Class modeling	√	√	√
Component and Deployment modeling	√		√
E-R Modeling	√	√	√

Acknowledgement

We would like to express our gratitude to the almighty God that let us to be able to successfully perform this project and also We express our sincere gratitude to our instructors Mr Derejaw, who have given timely advice as well as encouragement to our project and his continuous support in every phase of this project.

Finally, we would like to thank Bahir Dar University for providing the support that enabled us to do this project. Without their assistance, we would not have been able to gain this valuable experience..

LIST OF ACRONYMS

Acronym	Full Form
AI	Artificial Intelligence
API	Application Programming Interface
AES	Advanced Encryption Standard
CBT	Cognitive Behavioral Therapy
CRC	Class–Responsibility–Collaboration
CRUD	Create, Read, Update, Delete
DB	Database
ER	Entity Relation
FDA	Food and Drug Administration
GUI	Graphical User Interface
HTTP	Hypertext Transfer Protocol
HTTPS	Hypertext Transfer Protocol Secure
ICT	Information and Communication Technology
IDE	Integrated Development Environment
IT	Information Technology
JWT	JSON Web Token
PHI	Protected Health Information
RBAC	Role-Based Access Control
REST	Representational State Transfer
TLS	Transport Layer Security
UML	Unified Modeling Language
UI	User Interface
UX	User Experience

List of Figures

Figure 1: Use case diagram.....	14
Figure 2: UI prototype	25
Figure 3: State chart diagram for signup.....	26
Figure 4: State chart diagram for login	27
Figure 5: State chart diagram for Therapy Session Scheduling.....	27
Figure 6: State chart diagram Payment for Therapy Sessions	28
Figure 7: State chart diagram for Exposure Therapy	28
Figure 8: State chart diagram for CBT Exercises	28
Figure 9: State chart diagram for Mood and Progress Tracking	29
Figure 10: State chart diagram for Profile Management	29
Figure 11: Activity Diagram for sign up	30
Figure 12: Activity Diagram for login	30
Figure 13: Activity Diagram Profile Management	30
Figure 14: Activity Diagram for User Management(admin)	30
Figure 15: Activity Diagram for Anxiety Assessment	31
Figure 16: Activity Diagram Therapy Session Scheduling.....	31
Figure 17: Activity Diagram for Payment for Therapy Sessions.....	32
Figure 18: Activity Diagram for CBT Exercises	33
Figure 19: Activity Diagram for Exposure Therapy	33
Figure 20: Activity Diagram for logout	34
Figure 21: Activity Diagram for Mood and Progress Tracking	34
Figure 22: Sequence Diagram for sign up	35
Figure 23: Sequence Diagram for login.....	35
Figure 24: Sequence Diagram Therapy Session Scheduling	36
Figure 25: Sequence Diagram Payment for Payment	36
Figure 26: Sequence Diagram for AI Assistant.....	37
Figure 27: Sequence Diagram Profile Management	37
Figure 28: Sequence Diagram Therapist profile Management	38
Figure 29: Sequence Diagram Anxiety Assessment	38
Figure 30: Sequence Diagram CBT Exercises.....	39
Figure 31: Sequence Diagram Mood and Progress Tracking	39
Figure 32: Sequence Diagram Exposure Therapy	40
Figure 33: Sequence Diagram for logout.....	40
Figure 34: Analysis Class Model Diagram	41
Figure 35: Conceptual Model Class Diagram.....	48
Figure 36: patient UI.....	50
Figure 37 :System Administrator UI.....	51
Figure 38: Three-tier architecture model	57

Figure 39: Subsystem Model for Anxiety and phobia Treatment System	59
Figure 40: Component Diagram for Anxiety and phobia Treatment System	63
Figure 41: Deployment Diagram for Anxiety and Phobia Treatment System	66
Figure 42: Class Model	68
Figure 43: E-R Diagram	79
Figure 44: Patient (Mobile app) UI	79
Figure 45: Admin UI	80

List of Tables

Table 1: Description of <i>Functional Requirement</i>	13
Table 2: Description of Sign Up use case	15
Table 3: Description of Login use case	16
Table 4: Description of Profile Management use case.....	17
Table 5: Description of Anxiety Assessment use case	18
Table 6: Description of Therapy Session Scheduling use case	18
Table 7: Description of CBT Exercises use case	19
Table 8: Description of Exposure Therapy use case	20
Table 9: Description of Mood and Progress Tracking use case	20
Table 10: Description of Emergency Support use case.....	21
Table 11: Description of User Management use case	21
Table 12: Description of System Monitoring and Maintenance use case	22
Table 13: Description of Logout use case	22
Table 14: Description of Payment for Therapy Sessions use case.....	23
Table 15: Description of Business Rule.....	25
Table 16: change cases description	49
Table 17: Database Entity Table	77
Table 18: Access control matrix.....	82

Table of Contents

Contents	Page no
Declaration.....	ii
Roles and Responsibilities of the Group Members.....	iii
Acknowledgement	iv
List of acronyms	v
List of Figures	vi
List of Tables.....	vii
Abstract	xi
CHAPTER ONE : INTRODUCTION.....	1
1.1 Background of the Study	1
1.2 Statement of the Problem.....	1
1.3 Objectives of the Study	3
1.3.1 General Objective	3
1.3.2 Specific Objectives	3
1.4 Methodology	4
1.4.1 Research Approach	4
1.4.2 Implementation Methodology.....	5
1.4.3 Testing and Validation Methodology	5
1.4.4 Deployment and Maintenance Strategy	5
1.5 Feasibility Study	6
1.6 Beneficiaries of the Project.....	7
1.7 Limitations of the Project.....	7
1.8 Scope of the Project	8
1.9 Organization of the Project	8
Chapter Two : System features	10
2.1 The Existing System	10
2.2 Proposed System	11
2.3 Requirement Analysis	12
2.3.1 Functional Requirements	12

2.3.2 System Use Case.....	14
2.3.3 Business Rule Documentation	23
2.3.4 User Interface Prototype	25
2.3.5 State Chart Diagram.....	26
2.3.6 Activity Diagram.....	30
2.3.7 Sequence diagram	35
2.3.8 Analysis Class Model.....	41
2.3.9 Logic Model.....	41
2.4 Non-Functional Requirements	45
2.5 System Requirements.....	46
2.5.1 Hardware Requirements.....	46
2.5.2 Software Requirements	47
2.6 Key Abstraction with CRC Analysis.....	48
2.6.1 Conceptual Modeling: Class Diagram	48
2.6.1 Conceptual modeling: Class diagram	49
2.6.3 User Interface Prototyping.....	50
Chapter Three: System Design	52
3.1 Architectural Design	55
3.1.1 Component Modeling	60
3.1.2 Deployment Modeling	63
3.2 Detail Design	67
3.2.1 Design class model	67
3.2.2 Entity-Relationship (E-R) Model.....	77
3.3 User Interface Design	79
3.4 Access Control and Security	80
References-----	84

Abstract

Anxiety disorders and phobias are among the most common mental health challenges affecting individuals worldwide. Traditional exposure therapy is effective but often limited by accessibility, cost, and patient reluctance. This project proposes a Therapy for Treating Anxiety and Phobias Using an Exposure Simulation System, which provides a controlled, gradual, and safe digital environment for exposure-based intervention. The system allows users to face fear-inducing stimuli through multimedia content such as images, videos, animations, and interactive simulations. The project focuses on system design and implementation rather than medical diagnosis, making it suitable as a computer science solution that supports mental well-being through technology.

CHAPTER ONE : INTRODUCTION

1.1 Background of the Study

Anxiety disorders and phobias are among the most widespread mental health conditions affecting individuals across different age groups and social backgrounds. People experiencing these conditions often face persistent fear, panic, stress, and avoidance behaviors that interfere with their daily routines, academic performance, professional responsibilities, and social interactions. In many cases, these challenges reduce quality of life and prevent individuals from functioning effectively in society.

Although the number of people affected by anxiety and phobias continues to increase, access to professional mental health care remains limited. Factors such as a shortage of trained therapists, high treatment costs, social stigma, and geographical barriers prevent many individuals from receiving timely support. In particular, people living in rural or underserved areas often struggle to access mental health services, leading to delayed diagnosis and prolonged suffering.

Advances in information and communication technology have created new opportunities to address these challenges. Digital health solutions, including online therapy platforms, enable patients and mental health professionals to interact remotely using web and mobile technologies. Such systems support psychological assessments, therapy sessions, progress monitoring, and secure storage of sensitive patient information, thereby improving efficiency and accessibility.

This project proposes the design and development of a Therapy Management System for Treating Anxiety and Phobias. The system is intended to provide a digital platform that connects patients, therapists, and administrators in a secure and organized environment. By reducing reliance on manual processes and in-person interactions, the system aims to enhance mental health service delivery while maintaining ethical standards and data security.

1.2 Statement of the Problem

Anxiety disorders and phobias represent a serious and growing mental health concern worldwide. These conditions affect not only individuals' emotional well-being but also their cognitive performance, social relationships, academic success, and productivity. Many people suffering from anxiety and phobias experience chronic fear, avoidance behaviors, reduced self-confidence, and social isolation, which can worsen over time if appropriate intervention is not provided.

Despite increased awareness of mental health issues, access to professional therapy remains insufficient. Mental health services are often concentrated in urban centers, leaving individuals in

rural and remote areas with limited or no access to qualified professionals. Long waiting times, high consultation fees, and shortages of trained therapists further restrict service availability. In addition, social stigma associated with mental illness discourages many individuals from seeking face-to-face therapy, as they fear judgment, discrimination, or loss of privacy.

Traditional mental health care systems rely heavily on in-person consultations and manual administrative processes. Patient records, therapy notes, and progress reports are frequently maintained on paper or in fragmented digital formats. This approach leads to inefficiencies such as misplaced records, duplicated information, difficulty in tracking long-term progress, and limited coordination among healthcare providers. Therapists may also find it challenging to manage appointments, monitor multiple patients consistently, and evaluate treatment outcomes systematically.

From a technological perspective, the lack of an integrated therapy management system results in underutilization of available digital resources. Existing platforms often lack scalability, interoperability, and robust security mechanisms. As a result, sensitive mental health data may be exposed to privacy risks, undermining patient trust. Weak authentication and access control further increase the likelihood of data misuse or unauthorized access.

Furthermore, many digital health systems are designed for general healthcare services and do not adequately address the specific requirements of anxiety and phobia treatment. These conditions require continuous monitoring, personalized therapy plans, behavioral tracking, and regular feedback. Without a structured and specialized digital system, it becomes difficult to provide consistent, patient-centered, and evidence-based mental health care.

Ethical considerations are also frequently overlooked in system design. Mental health applications must respect confidentiality, informed consent, emotional vulnerability, and cultural sensitivity. When these principles are not adequately integrated, digital solutions risk causing harm rather than support.

Therefore, the central problem addressed by this study is the absence of a secure, accessible, and human-centered digital therapy management system specifically designed for treating anxiety and phobias. There is a clear need for a platform that integrates psychological principles, ethical responsibility, and modern information technology to improve mental health care delivery.

1.3 Objectives of the Study

The objectives of this study are formulated to address the psychological, social, ethical, and technological challenges identified in the problem statement. The project seeks to combine human-centered mental health care principles with information technology solutions to enhance the accessibility, effectiveness, and management of therapy services for anxiety and phobias.

1.3.1 General Objective

The general objective of this study is to design and develop a secure, accessible, and user-focused digital therapy management system that supports the treatment of anxiety and phobias through the integration of psychological, ethical, and technological principles.

1.3.2 Specific Objectives

The specific objectives of this study are to:

1. **Improve Access to Mental Health Services:**
To design a digital platform that enables individuals suffering from anxiety and phobias to access professional therapy services regardless of geographical location, thereby reducing barriers related to distance, mobility, and availability of therapists.
2. **Reduce Social Stigma through Digital Interaction:**
To provide a private and confidential environment where patients can seek therapy discreetly, encouraging individuals who may avoid face-to-face sessions due to social stigma or fear of judgment.
3. **Support Psychological Self-Assessment and Early Identification:**
To incorporate structured self-assessment tools based on psychological questionnaires that help identify the severity of anxiety and phobia symptoms and support early intervention.
4. **Enhance Therapist Efficiency and Clinical Decision-Making:**
To develop tools that allow therapists to manage patient records, schedule sessions, document therapy notes, and monitor progress systematically, enabling evidence-based and consistent treatment decisions.
5. **Enable Personalized and Continuous Therapy Management:**
To support the creation and modification of individualized treatment plans based on patient progress, behavioral patterns, and therapeutic feedback over time.
6. **Ensure Ethical Handling of Sensitive Mental Health Data:**
To design and implement secure data storage, authentication, and access control

mechanisms that protect patient privacy, ensure confidentiality, and comply with ethical standards of mental health care.

7. Integrate Human-Centered User Interface Design:

To develop user interfaces that are empathetic, simple, and accessible, considering the emotional and cognitive states of users experiencing anxiety or phobia.

8. Support Progress Tracking and Outcome Evaluation:

To provide analytical tools and visual reports that help both patients and therapists understand therapy outcomes, behavioral changes, and improvement trends over time.

9. Facilitate Administrative Oversight and System Governance:

To enable administrators to manage users, verify therapist credentials, monitor system usage, and ensure compliance with operational and ethical policies.

10. Promote Interdisciplinary Application of Humanities and IT:

To demonstrate how concepts from psychology, ethics, and social sciences can be effectively integrated with information technology to create meaningful digital health solutions.

1.4 Methodology

1.4.1 Research Approach

To develop a simple, effective, and user-friendly system for treating phobia and anxiety, different methods were used to collect system requirements and understand user needs.

Interviews were conducted with individuals experiencing phobia and anxiety, as well as mental health professionals such as therapists and counselors. These interviews helped identify common problems faced by users, their expectations, and the type of support they need from the system. Users emphasized the need for features such as anxiety self-assessment, guided relaxation exercises, therapy session scheduling, and AI chatbot support. Mental health professionals provided guidance to ensure that the system follows basic therapeutic principles and ethical practices.

Review of existing systems and studies was carried out by analyzing similar mental health applications, research articles, and case studies related to anxiety and phobia treatment. This review helped understand how current systems work and identify areas where improvements can be made, such as better user interaction, easy accessibility, and continuous mental health support.

Review of mental health guidelines and reports from recognized health organizations and national health authorities was conducted to ensure that the system aligns with recommended mental health practices. These guidelines helped shape system features that promote mental well-being, early intervention, and awareness about anxiety and phobia.

After collecting the requirements, the information was organized into functional and non-functional requirements based on importance and system objectives. This organization helped in designing the system models and defining the main features of the application. The final requirements were then used to create user stories and functional specifications, which guided the system design and development process

1.4.2 Implementation Methodology

The system implementation follows a modular development approach:

- Development of frontend components for different user roles
- Implementation of backend services for business logic and communication
- Integration of database operations for data storage and retrieval
- Implementation of security features and access control
- Integration of communication tools for online therapy sessions

Each module is developed and tested independently before system integration.

1.4.3 Testing and Validation Methodology

Tools and Technologies Used

Flutter: The app will be developed using Flutter as the primary programming framework.

Flutter is well-suited for mobile app development due to its cross-platform capabilities, allowing the app to run on both Android and iOS devices, maximizing its reach.

Dart: Flutter uses Dart as the programming language, which is optimized for performance and user interface design. Dart will be used for coding the app's logic and UI elements.

Node: is the primary programming framework backend. Since it is well developed for our interactive and complex system.

Database PostgreSQL: For storage, such as health tracking information, user profiles, and reminders. This allows for efficient access to the app's feature.

Firebase: For cloud storage and synchronization (teleconsultation, user feedback)

1.4.4 Deployment and Maintenance Strategy

The system is deployed on a secure server environment. Maintenance includes:

- Regular system updates
- Security patching
- Data backup and recovery
- Continuous improvement based on user feedback

1.5 Feasibility Study

This feasibility study evaluates the practicality of developing and implementing the proposed Therapy Management System for Treating Anxiety and Phobias. The analysis considers technical, economic, operational, legal, and ethical aspects.

Technical Feasibility

The system is technically feasible because it relies on commonly available web, mobile, backend, and database technologies. No specialized hardware is required, and the system architecture is scalable and reliable.

Economic Feasibility

The system is cost-effective as it reduces expenses related to physical therapy visits, paperwork, and administration. Development and maintenance costs are reasonable within an academic or institutional setting.

Operational Feasibility

The system is user-friendly and designed using a human-centered approach. Patients, therapists, and administrators can use the system with minimal training.

Legal Feasibility

The system follows general data protection and privacy principles by ensuring secure handling of sensitive mental health information and user consent.

Ethical Feasibility

Ethical considerations such as confidentiality, psychological safety, and respect for human dignity are integrated into the system design.

Schedule Feasibility

The project can be completed within the allocated academic timeline using a phased development approach.

1.6 Beneficiaries of the Project

The proposed Therapy Management System for Treating Anxiety and Phobias is significant to individuals, institutions, and society by improving access to mental health services through digital technology.

Beneficiaries

- **Patients:**
Patients benefit from easy access to therapy services, increased privacy, reduced stigma, and continuous support regardless of location.
- **Therapists:**
Therapists benefit from efficient management of patient records, therapy sessions, and progress tracking, which improves service delivery and clinical decision-making.
- **Administrators and Institutions:**
Healthcare institutions and system administrators benefit from improved system control, secure data management, and reliable reporting.
- **Society:**
Society benefits from increased mental health awareness, early intervention, and reduced social and economic impacts of untreated anxiety and phobias.

From a humanities perspective, the project promotes human dignity, emotional well-being, and ethical mental health care. From an information technology perspective, it demonstrates how digital systems can be used to manage sensitive healthcare data securely and efficiently.

Overall, the project contributes to improved mental health service delivery, reduced barriers to care, and the effective integration of humanities and information technology in healthcare solutions.

1.7 Limitations of the Project

Despite the benefits of the proposed Therapy Management System for Treating Anxiety and Phobias, the project has certain limitations:

- The system requires stable internet connectivity, which may limit access for users in areas with poor network infrastructure.
- The system does not replace emergency or crisis mental health services and is not suitable for severe psychiatric conditions requiring immediate intervention.
- The effectiveness of the system depends on user honesty and active participation during self-assessment and therapy sessions.

- The system is limited to the treatment and management of anxiety and phobias and does not cover other mental health disorders.
- Technical issues such as system downtime or device incompatibility may affect service availability.

1.8 Scope of the Project

The scope of this project focuses on the design and development of a digital Therapy Management System for Treating Anxiety and Phobias. The system is intended to support online mental health therapy by integrating psychological principles with information technology solutions.

The project covers user registration and authentication for patients, therapists, and administrators. It includes digital self-assessment tools for evaluating anxiety and phobia symptoms, online therapy session scheduling, and communication through chat or video platforms. The system also supports progress tracking, therapy reports, and feedback mechanisms to monitor patient improvement over time.

Additionally, the project includes secure data storage, role-based access control, and administrative functions such as user management and system monitoring. The system is limited to software-based solutions and does not include medication prescription, emergency mental health response, or physical clinical services.

Overall, the scope is confined to the management and support of anxiety and phobia therapy through a secure and user-friendly digital platform.

1.9 Organization of the Project

This project is organized into three main chapters, each focusing on a key aspect of the design and development of the Therapy Management System for Treating Anxiety and Phobias. The structure of the project is described as follows:

Chapter One: Introduction

This chapter presents an overview of the project, including the background of the study, statement of the problem, objectives, methodology, feasibility study, scope, limitations, and significance of the project. It provides the foundation and context for understanding the need for a digital therapy management system for anxiety and phobias.

Chapter Two: System Features and Requirements

This chapter describes the main features and functional components of the system. It explains the roles of patients, therapists, and administrators, and outlines system functionalities such as self-assessment, therapy session scheduling, online therapy, progress tracking, and reporting. Functional and non-functional requirements are also discussed to define how the system should operate.

Chapter Three: System Design

This chapter focuses on the design of the system and presents the overall system architecture. It includes database design, user interface layout, and system workflow models. Diagrams such as use case diagrams, class diagrams, and data flow diagrams are used to illustrate how system components interact and how the system is structured.

References

A comprehensive list of all sources cited throughout the project

Chapter Two : System features

2.1 The Existing System

This section describes the current system used in therapy centers for treating anxiety and phobias. The existing system is primarily manual, relying on physical interactions between patients and therapists. There is no centralized digital platform or integrated application to manage patient information, appointments, assessments, or therapy progress.

How the Existing System Functions

1. Patient Registration

- Patients visit the clinic in person to register.
- They fill out paper-based registration forms with personal information and medical history.
- Receptionists manually store these records in physical files or, in some cases, basic spreadsheets.

2. Appointment Scheduling

- Patients book therapy sessions by phone calls or visiting the clinic.
- Therapists' availability is checked using manual calendars or notebooks.
- Appointments are written down manually, and reminders are often communicated verbally or via calls.

3. Therapy Sessions

- Therapy sessions are conducted face-to-face by therapists.
- Notes about patient progress and session outcomes are written by hand.
- There is no digital record of sessions, making it difficult to review past sessions efficiently.

4. Assessment and Progress Tracking

- Anxiety and phobia assessments are conducted using printed questionnaires.
- Therapists manually evaluate the results and provide treatment recommendations.
- Progress tracking depends on therapists' memory and handwritten notes, without automated scoring or visual analysis.

5. Report Management

- Therapy reports, session summaries, and assessment results are stored in physical files.
- Retrieving old reports is time-consuming and prone to error.
- Data security is weak as paper files are susceptible to loss or damage.

Drawbacks of the Existing System

- **Time-consuming:** Manual processes for registration, booking, and reporting slow down operations.
- **Data loss risk:** Paper records are vulnerable to damage or misplacement.
- **Limited accessibility:** Patients must physically visit the clinic; remote therapy is not possible.
- **Inaccurate progress tracking:** Manual evaluation leads to inconsistent records.

- **Poor communication:** Limited channels for patients to interact with therapists outside clinic hours.

The limitations of the current system highlight the need for a digital therapy management system that automates patient registration, session scheduling, assessments, progress tracking, and reporting while improving accessibility, efficiency, and data security.

2.2 Proposed System

The proposed system is a digital therapy management platform designed to address the limitations of the existing manual system. It integrates patient registration, appointment scheduling, therapy sessions, assessments, progress tracking, and reporting into a centralized and automated system, providing both patients and therapists with a user-friendly interface.

Key Features and Solutions

1. **Digital Patient Registration**
 - Patients can register online through the platform or mobile app.
 - Personal information, medical history, and therapy preferences are securely stored in the system.
 - Eliminates paper forms and reduces errors in data entry.
2. **Automated Appointment Scheduling**
 - Patients can view therapist availability and book sessions online.
 - Automatic reminders are sent via email or app notifications.
 - Prevents double bookings and saves time for both patients and therapists.
3. **Therapy Session Management**
 - Therapists can conduct and manage sessions efficiently.
 - Session notes and outcomes are recorded digitally for easy retrieval.
 - Option for remote sessions via integrated online tools can improve accessibility.
4. **Digital Assessments and Progress Tracking**
 - Anxiety and phobia assessments are digitized.
 - Automatic scoring and visual progress charts help both patients and therapists track improvement over time.
 - Facilitates data-driven therapy decisions and personalized treatment plans.
5. **Report Generation and Management**
 - Therapy reports, assessment results, and session summaries are generated automatically.
 - Reports are stored securely in the system, allowing easy access and retrieval.
 - Improves data security and ensures patient confidentiality.
6. **Optional AI-Assisted Support**
 - AI can provide patients with self-help guidance, relaxation exercises, or preliminary assessments.
 - Clearly acts as a support tool, not replacing the therapist.

Benefits of the Proposed System

- **Efficiency:** Reduces manual work and administrative burden.
- **Accuracy:** Digital storage prevents errors and data loss.
- **Accessibility:** Patients can access therapy services remotely.
- **Improved Tracking:** Therapists can monitor patient progress using charts and digital reports.
- **Enhanced Communication:** Facilitates secure communication between patients and therapists outside clinic hours.

2.3 Requirement Analysis

2.3.1 Functional Requirements

Functional requirements describe what the system must do to fulfill its objectives. Each requirement is uniquely identified and prioritized based on its importance to the system.

ID	Functional Requirement	Description	Priority
FREQ-1	User Registration	Allow patients, therapists, and admins to create accounts and store personal and professional details securely.	High
FREQ-2	User Login & Authentication	Enable users to securely log in with email/username and password. Include password recovery.	High
FREQ-3	Patient Profile Management	Patients can view and update personal information, therapy history, and assessment results.	High
FREQ-4	Therapist Profile Management	Therapists can manage their profile, specialization, available schedule, and patient assignments.	High
FREQ-5	Admin Dashboard	Admins can manage all users, verify therapist credentials, and monitor overall system usage.	High
FREQ-6	Appointment Scheduling	Patients can book therapy sessions based on therapist availability; system prevents double-booking.	High
FREQ-7	Appointment Notifications	Automatic reminders for patients and therapists via email or app notifications before scheduled sessions.	Medium
FREQ-	Therapy Session	Therapists can record session notes, upload session	High

8	Management	materials, and track patient attendance.	
FREQ-9	Digital Assessments	Conduct anxiety and phobia questionnaires digitally, store responses, and automatically calculate results.	High
FREQ-10	Progress Tracking & Reports	Generate visual charts of patient progress over time; allow therapist feedback and recommendations.	High
FREQ-11	AI-Assisted Support	Optional AI module provides guidance, relaxation exercises, and preliminary assessments for patients.	Medium
FREQ-12	Secure Data Storage	All patient data, assessments, session notes, and reports are securely stored with access control.	High
FREQ-13	Search and Retrieval	Allow patients, therapists, and admins to search for past sessions, reports, or assessment results easily.	Medium
FREQ-14	Remote Access	Support secure remote login for patients and therapists to access the system from anywhere.	High
FREQ-15	Reporting & Analytics	Admins can generate system usage reports and statistical analysis on therapy outcomes.	Medium

Table 1: Description of *Functional Requirement*

2.3.2 System Use Case

2.3.2.1 Use Case Diagram

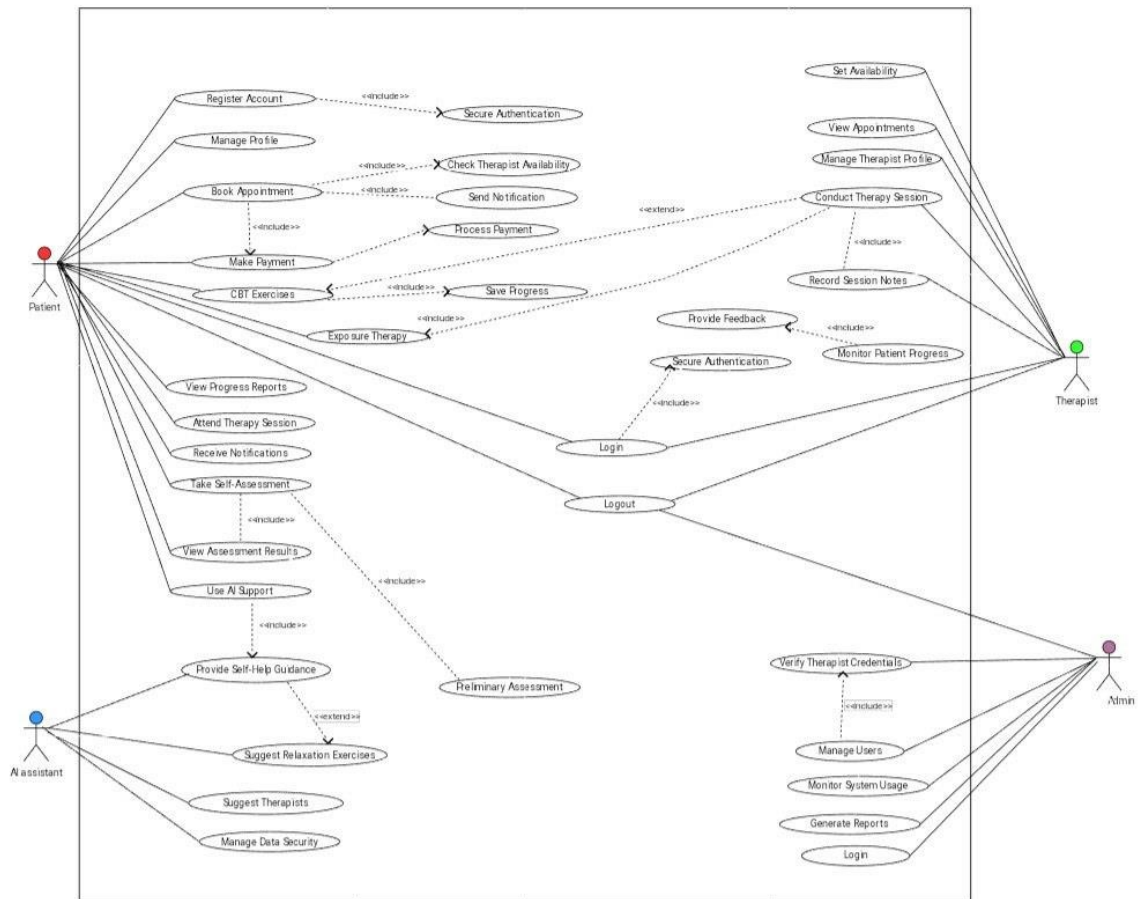


Figure 1: Use case diagram

2.3.2.2 Use Case Documentation

Use Case 1

Name	Sign Up
Actor	Patient
Description	Allows new users (patients) to register on the therapy platform by providing required personal and account details.
Pre-condition	The patient is not registered in the system.
Post-condition	The patient account is successfully created and stored in the system.
Priority	High
Basic Course of Action	1. The patient opens the application 2. The patient selects the Sign Up option. 3. The system displays the registration form. 4. The patient enters required details such as name, email, and password. 5. The system validates the entered information. 6. The system creates a new patient account. 7. The system displays a success message and redirects the patient to the login page.
Alternate Course of Action	1. Patient enters invalid or duplicate information → The system displays an error message and requests correction. 2. System encounters technical issues → The system displays an error message and asks the patient to try again later.

Table 2: Description of Sign Up use case

Use Case 2

Name	Login
Actor	Patient, Therapist, Admin
Description	Allows registered users to log in to the system using valid credentials.
Pre-condition	User account exists and is active.
Post-condition	User is successfully logged into the system and redirected to their dashboard.
Priority	High
Basic Course of Action	1. User navigates to the Login page. 2. User enters email and password. 3. The system verifies the credentials. 4. User is redirected to the appropriate dashboard.
Alternate Course of Action	1. Invalid credentials entered → The system displays an error message. 2. Multiple failed login attempts → The system temporarily locks the account. 3. Network issue occurs → The system displays a retry message.

Table 3: Description of Login use case

Use Case 3

Name	Profile Management
Actor	Patient, Therapist
Description	Allows users to view and update their profile information.
Pre-condition	User is logged in.
Post-condition	Profile information is updated successfully.

Priority	Medium
Basic Course of Action	1. User navigates to the Profile section. 2. User edits personal details. 3. User saves changes. 4. The system updates the profile information.
Alternate Course of Action	1. Unauthorized modification attempt → The system denies access. 2. System error occurs → The user is notified to try again later.

Table 4: Description of Profile Management use case

Use Case 4

Name	Anxiety Assessment
Actor	Patient
Description	Allows patients to take anxiety and phobia assessment tests.
Pre-condition	Patient is logged in.
Post-condition	Assessment results are saved and analyzed.
Priority	High
Basic Course of Action	1. Patient selects Anxiety Assessment. 2. System displays questionnaire. 3. Patient answers questions. 4. System analyzes responses. 5. System displays assessment results.
Alternate Course of Action	1. Patient exits before completion → Partial progress is saved.

	2. Technical error occurs → System prompts retry.
--	---

Table 5: Description of Anxiety Assessment use case

Use Case 5

Name	Therapy Session Scheduling
Actor	Patient
Description	Allows patients to schedule therapy sessions with therapists.
Pre-condition	Patient is logged in Therapist is available.
Post-condition	Therapy session is scheduled successfully.
Priority	High
Basic Course of Action	1. Patient selects Schedule Session. 2. Patient chooses therapist and preferred time. 3. System checks therapist availability. 4. Session is scheduled. 5. Notification is sent to patient and therapist.
Alternate Course of Action	1. No available slots → System suggests alternative times. 2. Therapist cancels session → Patient is notified and rescheduling is offered.

Table 6: Description of Therapy Session Scheduling use case

Use Case 6

Name	CBT Exercises
Actor	Patient
Description	Allows patients to perform Cognitive Behavioral Therapy exercises.
Pre-condition	Patient is logged in.
Post-condition	Exercise progress is recorded.
Priority	High
Basic Course of Action	1. Patient selects CBT Exercises. 2. System displays exercise modules. 3. Patient completes exercises. 4. System saves progress.
Alternate Course of Action	1. Patient exits mid-session → Progress is saved automatically. 2. Content loading error → Retry option is displayed.

Table 7: Description of CBT Exercises use case

Use Case 7

Name	Exposure Therapy
Actor	Patient, Therapist
Description	Allows patients to perform guided exposure therapy under therapist supervision.
Pre-condition	Patient is logged in. Therapist approval is obtained.
Post-condition	Exposure therapy progress is updated.
Priority	High
Basic Course of Action	1. Therapist assigns exposure plan.

	2. Patient starts exposure session. 3. System monitors progress. 4. Therapist reviews results.
Alternate Course of Action	1. Patient experiences distress → Emergency support is triggered. 2. System interruption occurs → Session is paused and saved.

Table 8: Description of Exposure Therapy use case

Use Case 8

Name	Mood and Progress Tracking
Actor	Patient
Description	Allows patients to track daily mood and therapy progress.
Pre-condition	Patient is logged in.
Post-condition	Mood and progress data are stored and analyzed.
Priority	Medium
Basic Course of Action	1. Patient logs mood status. 2. System records data. 3. Progress charts are updated.
Alternate Course of Action	1. Patient skips entries → System sends reminders. 2. Data storage error → Retry message displayed.

Table 9: Description of Mood and Progress Tracking use case

Use Case 9

Name	Emergency Support
Actor	Patient
Description	Allows patients to access immediate emergency mental health support.
Pre-condition	Patient is logged in.

Post-condition	Emergency support is activated.
Priority	High
Basic Course of Action	1. Patient clicks Emergency Help. 2. System connects to emergency contacts. 3. Therapist/Admin is alerted.
Alternate Course of Action	1. Network failure occurs → Emergency contact numbers are displayed.

Table 10: Description of Emergency Support use case

Use Case 10

Name	User Management
Actor	Admin
Description	Allows the admin to manage patient and therapist accounts.
Pre-condition	Admin is logged in.
Post-condition	User records are updated successfully.
Priority	High
Basic Course of Action	1. Admin views user list. 2. Admin adds, updates, or deletes users. 3. Changes are saved.
Alternate Course of Action	1. Unauthorized action attempted.→ Access denied. 2. System error occurs.→ Admin is notified.

Table 11: Description of User Management use case

Use Case 11

Name	System Monitoring and Maintenance
Actor	Admin
Description	Allows admin to monitor system performance and perform maintenance tasks.

Pre-condition	Admin is logged in.
Post-condition	System performance is maintained.
Priority	Medium
Basic Course of Action	1. Admin checks system logs. 2. Admin performs maintenance tasks. 3. System status is updated.
Alternate Course of Action	Critical system failure occurs.→ Emergency maintenance is initiated.

Table 12: Description of System Monitoring and Maintenance use case

Use Case 12

Name	Logout
Actor	Patient, Therapist, Admin
Description	Allows users to securely log out of the system.
Pre-condition	User is logged in.
Post-condition	User session is terminated.
Priority	Medium
Basic Course of Action	1. User selects Logout. 2. System ends the session.
Alternate Course of Action	System error during logout.→ Retry option is displayed.

Table 13: Description of Logout use case

Use Case 13

Name	Payment for Therapy Sessions
Actor	Patient
Description	Allows patients to make payments for scheduled therapy sessions before attending. Payment ensures session confirmation and access to therapy services. Methods include credit/debit card, mobile payment, or other

	integrated gateways.
Pre-condition	•Patient is logged in. •Therapy session has been selected and is pending confirmation.
Post-condition	• Payment is successfully processed, and the session is confirmed. • Payment failure prevents session confirmation; patient must retry or cancel. •Refunds processed if sessions are canceled within allowed timeframe.
Priority	High
Basic Course of Action	1. Patient selects a therapy session. 2. System calculates total session cost. 3. Patient chooses a payment method. 4. Patient submits payment. 5. System processes the payment. 6. If payment succeeds: • System confirms the therapy session. • Notifications are sent to patient and therapist. • Payment status updated to Completed. 7. If payment fails: • System displays error message. • Patient can retry payment or cancel session.
Alternate Course of Action	1. Payment fails due to insufficient funds, network error, or invalid details.→ System prompts patient to retry or cancel. 2. Payment is refunded for canceled sessions within allowed timeframes.→ System updates status to Refunded and notifies patient. 3. Payment gateway temporarily unavailable.→ System informs patient and allows retry when available.

Table 14: Description of Payment for Therapy Sessions use case

2.3.3 Business Rule Documentation

Identifier	Name	Description	Related Use Legal Case(s)	Reference
BR-001	Unique Account Credentials	Each user must register with a unique email address. Duplicate registrations are not allowed.	Sign Up, Login	Platform Registration Policy
BR-002	Authentication Control	Only registered and active users (Patient, Therapist, Admin) can log into the system.	Login	System Security Policy
BR-003	Profile Access Control	Users can only view and edit their own profile information.	Profile Management	Access Control Policy
BR-004	Assessment Access Rule	Anxiety and phobia assessments are accessible only to logged-in patients.	Anxiety Assessment	Clinical Assessment Guidelines
BR-005	Therapist Availability Validation	Therapy sessions can only be scheduled if the selected therapist is available at the chosen time.	Therapy Session Scheduling	Scheduling Policy
BR-006	CBT Progress Tracking	The system must automatically save patient progress during CBT exercises.	CBT Exercises	Therapy Monitoring Policy
BR-007	Therapist Approval Requirement	Exposure therapy sessions must be approved and supervised by a licensed therapist.	Exposure Therapy	Mental Health Safety Standards
BR-008	Mood Data Retention	Mood and progress data must be stored securely and linked to the patient's therapy history.	Mood & Progress Tracking	Data Management Policy
BR-009	Emergency Escalation Protocol	If a patient triggers emergency support, the system must immediately alert the assigned therapist or admin.	Emergency Support	Emergency Response Guidelines
BR-010	User Administration Authority	Only admins are allowed to add, update, or delete patient and therapist accounts.	User Management	Administrative Policy
BR-011	System Maintenance Control	Only admins can perform system monitoring and maintenance activities.	System Monitoring & Maintenance	System Operations Policy

BR-012	Secure Session Termination	All users must be securely logged out, and sessions must be terminated after logout.	Logout	Security Policy
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Table 15: Description of Business Rule

2.3.4 User Interface Prototype

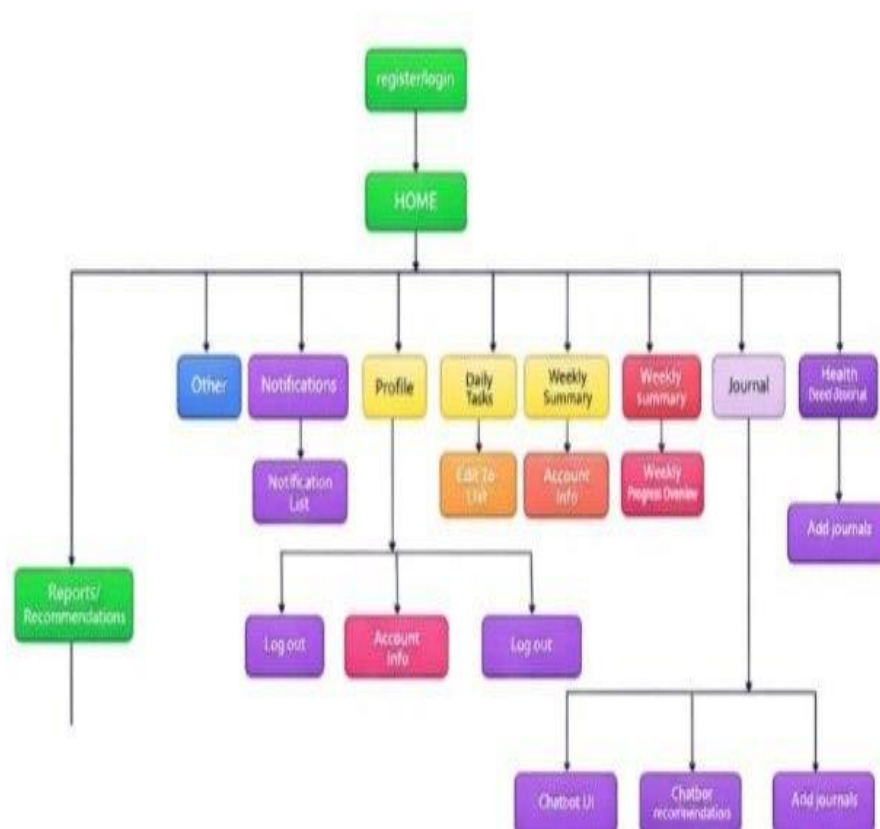


Figure 2: UI prototype

2.3.5 State Chart Diagram

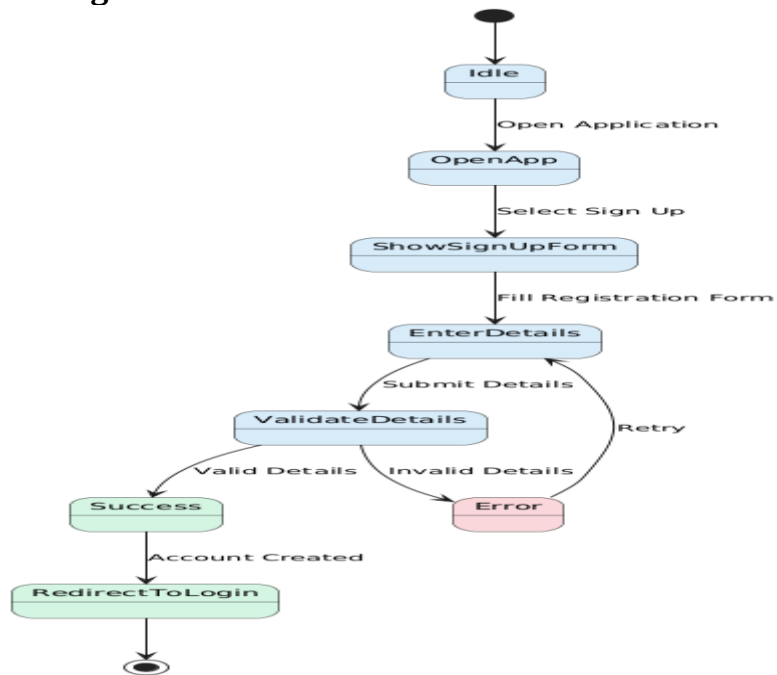


Figure 3: State chart diagram for signup

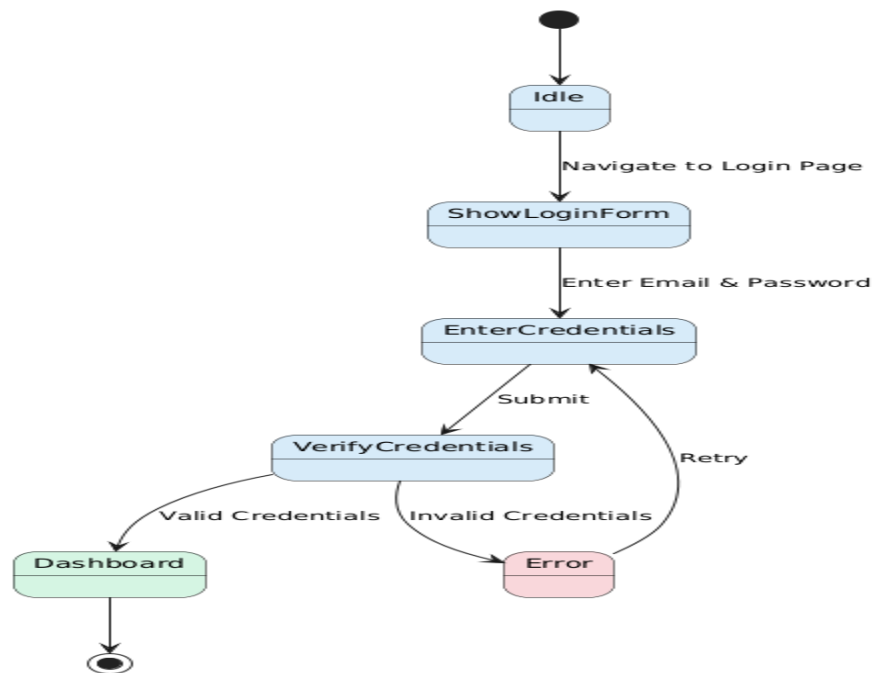


Figure 4: State chart diagram for login

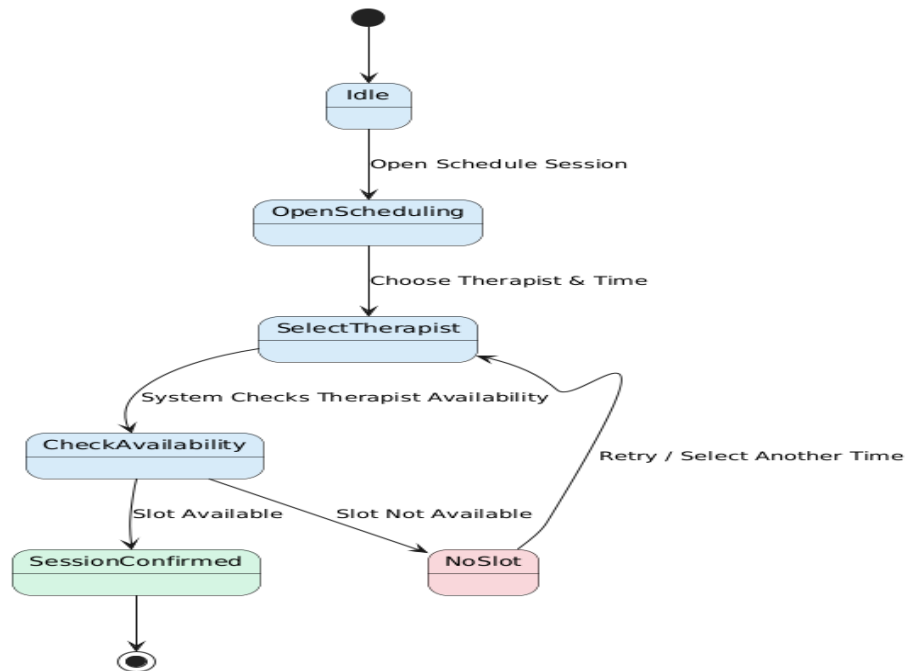


Figure 5: State chart diagram for Therapy Session Scheduling

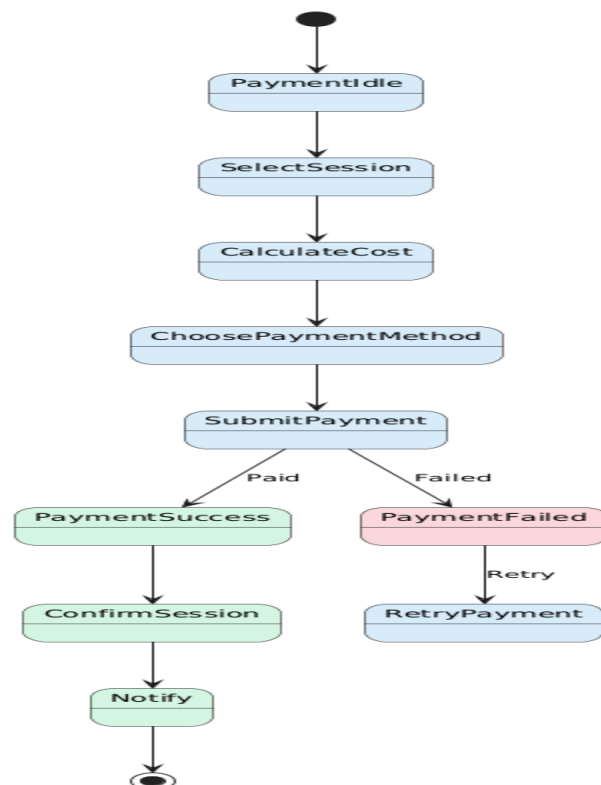


Figure 6: State chart diagram Payment for Therapy Sessions

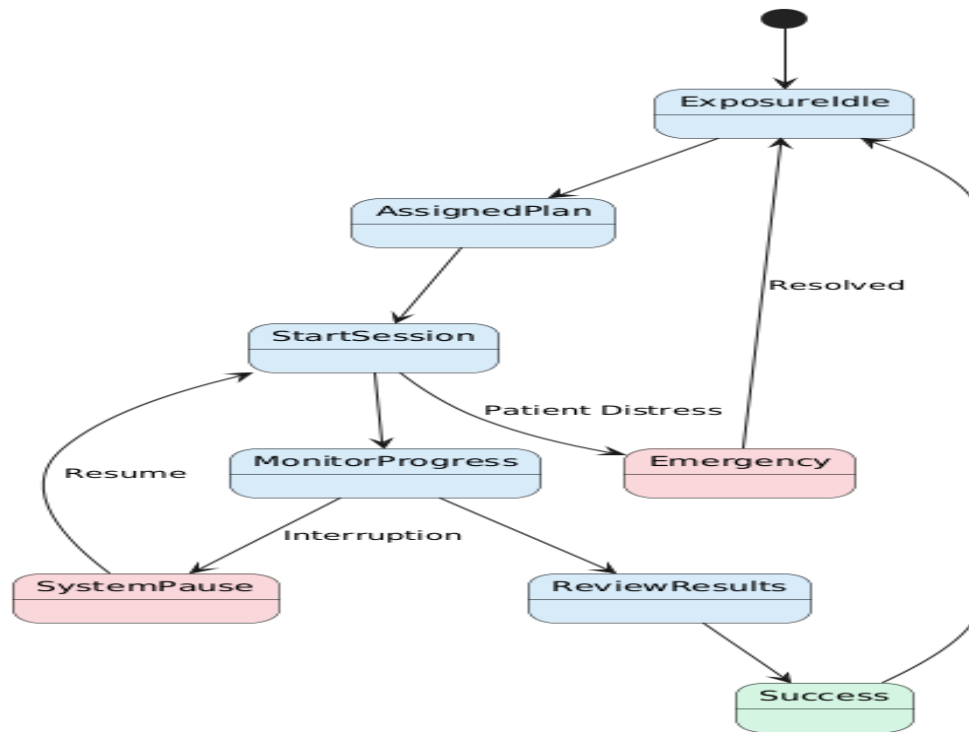


Figure 7: State chart diagram for Exposure Therapy

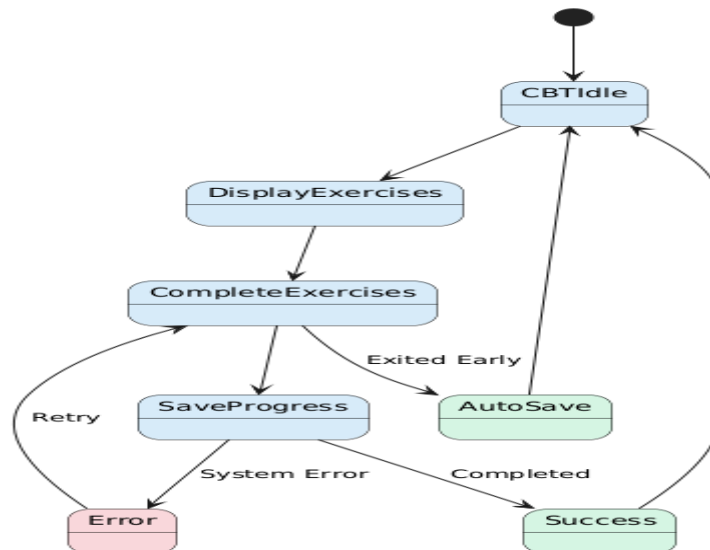


Figure 8: State chart diagram for CBT Exercises

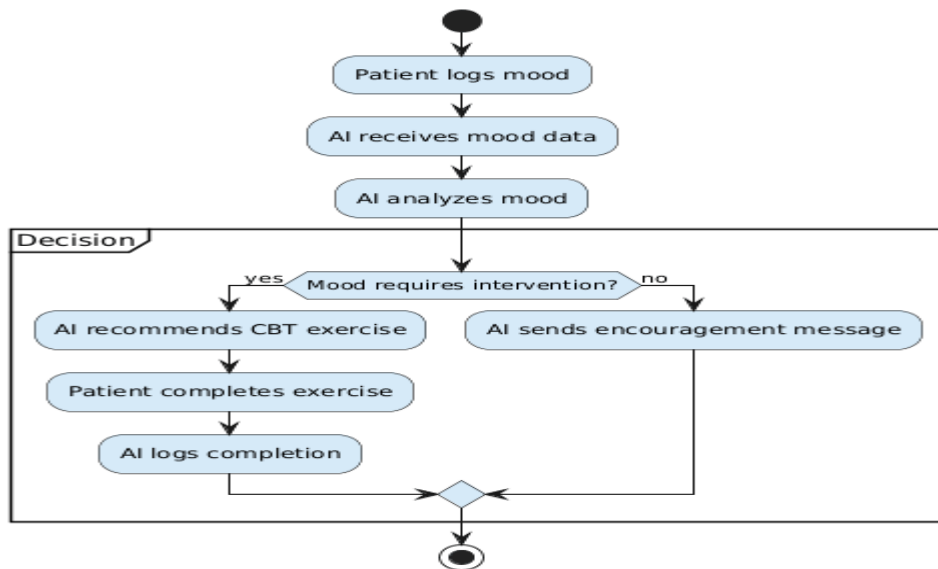


Figure 9: State chart diagram for Mood and Progress Tracking

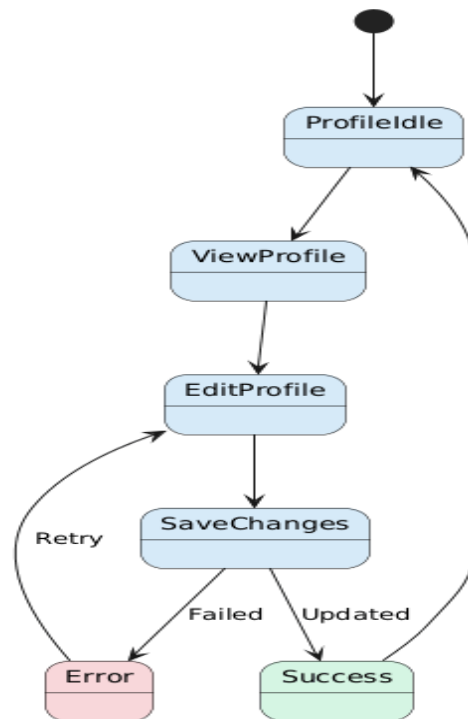


Figure 10: State chart diagram for Profile Management

2.3.6 Activity Diagram

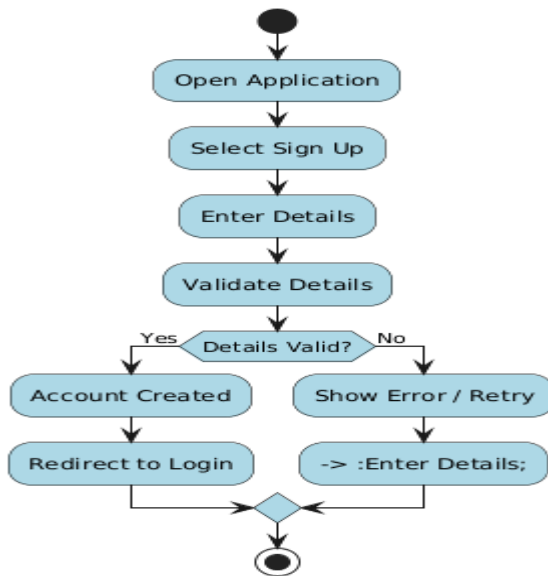


Figure 11: Activity Diagram for sign up

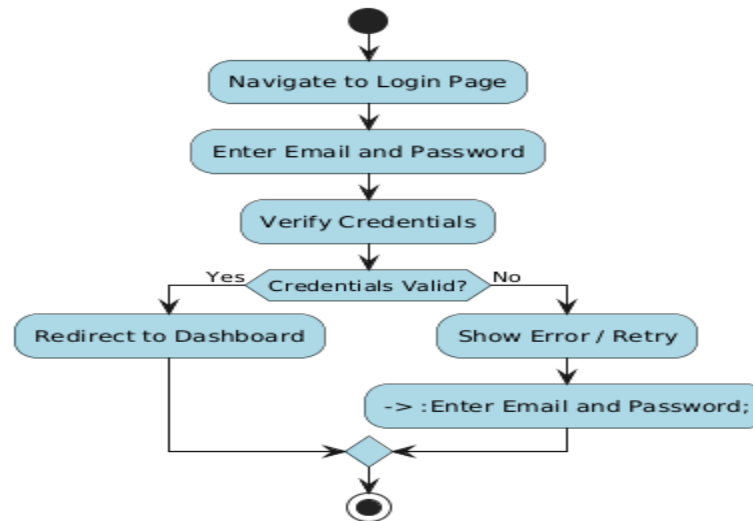


Figure 12: Activity Diagram for login

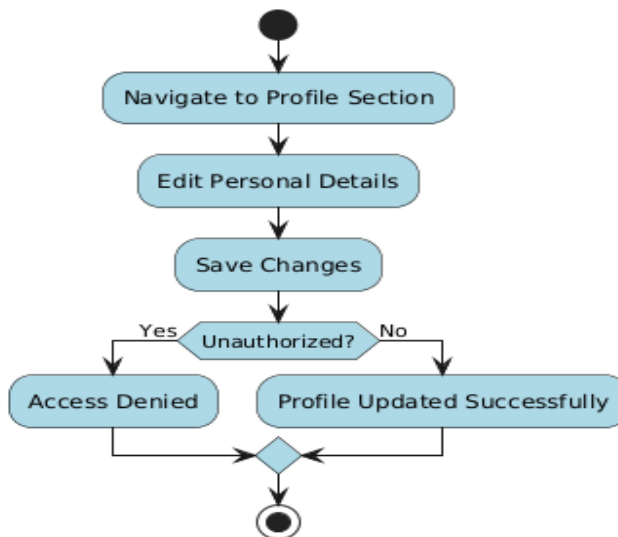


Figure 13: Activity Diagram Profile Management

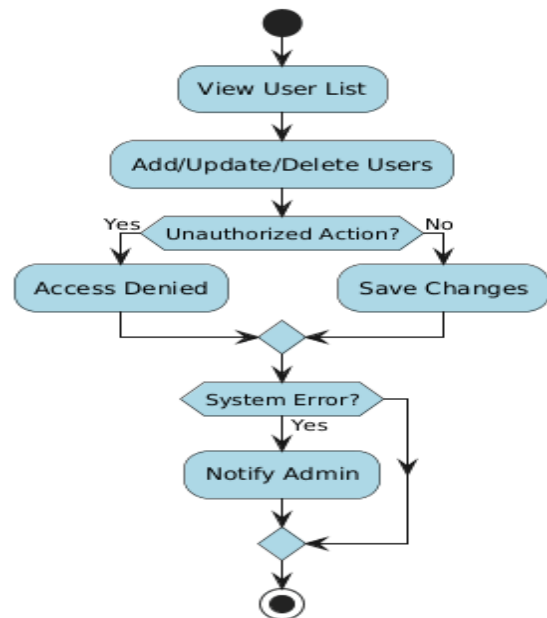


Figure 14: Activity Diagram for User Management(admin)

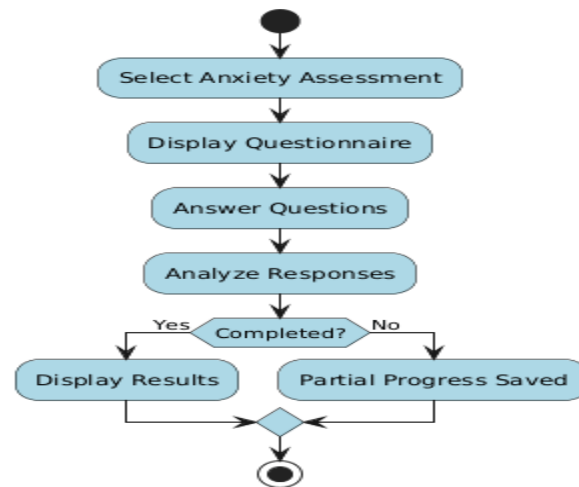


Figure 15: Activity Diagram for Anxiety Assessment

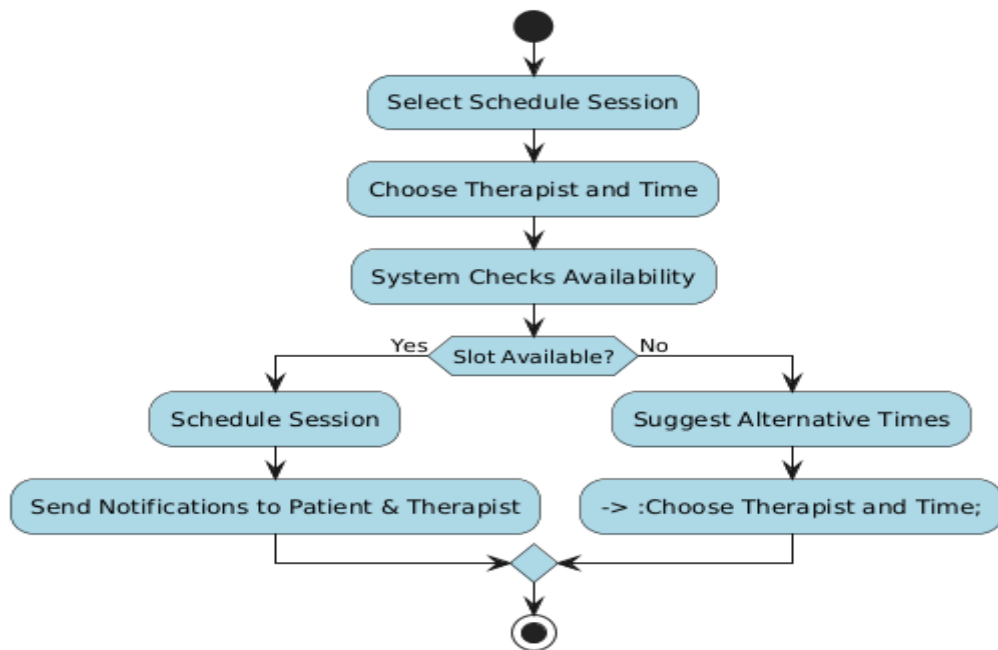


Figure 16: Activity Diagram Therapy Session Scheduling

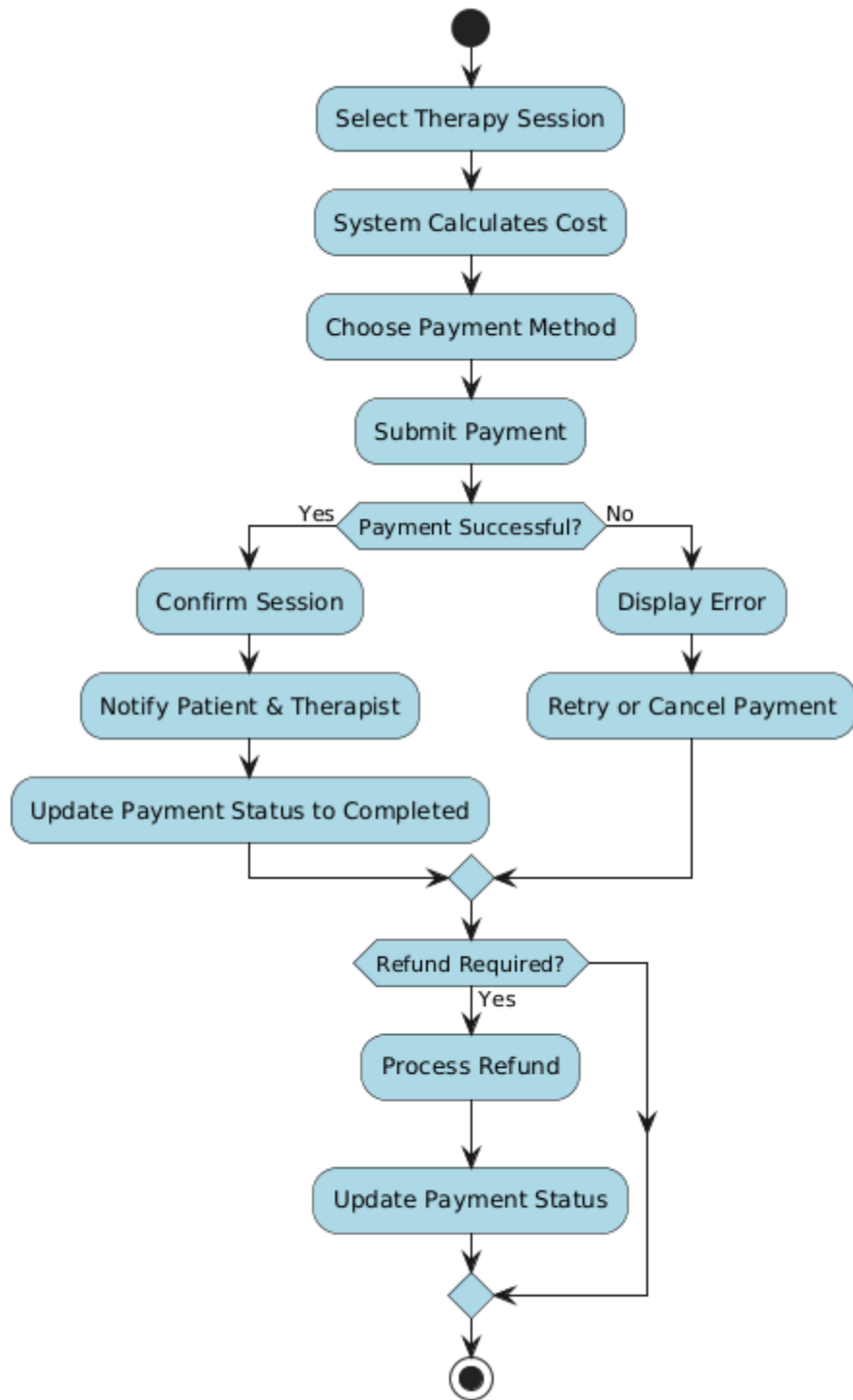


Figure 17: Activity Diagram for Payment for Therapy Sessions

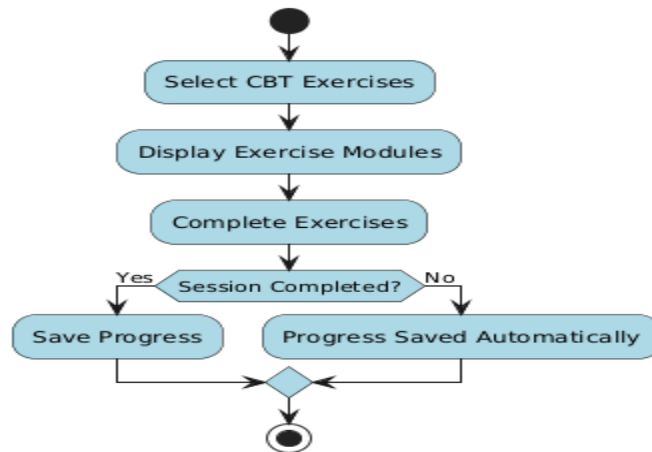


Figure 18: Activity Diagram for CBT Exercises

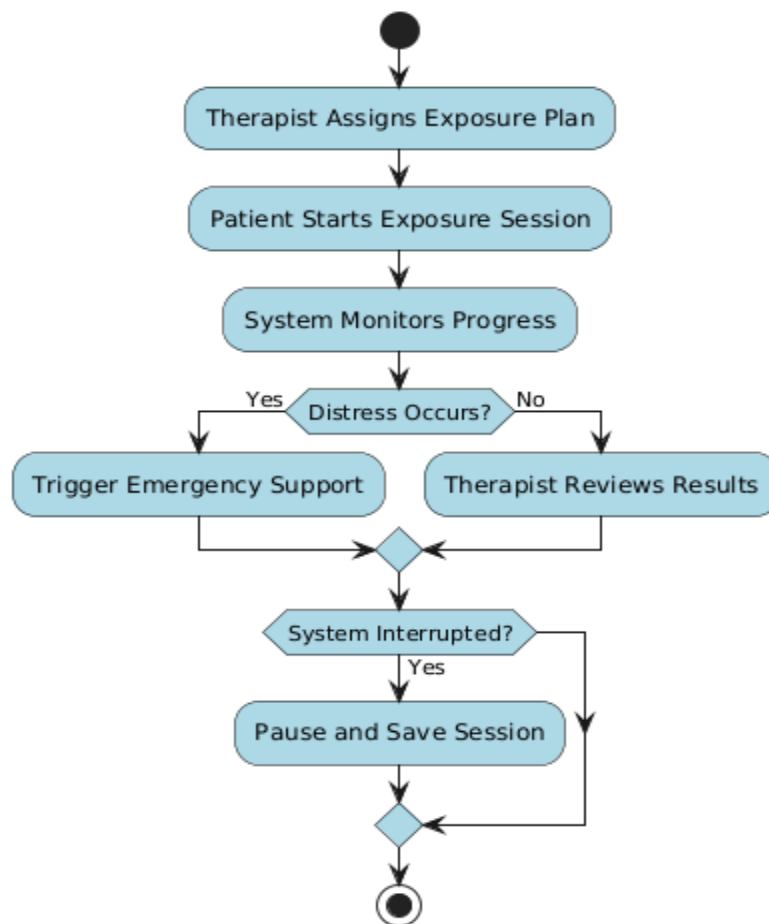


Figure 19: Activity Diagram for Exposure Therapy

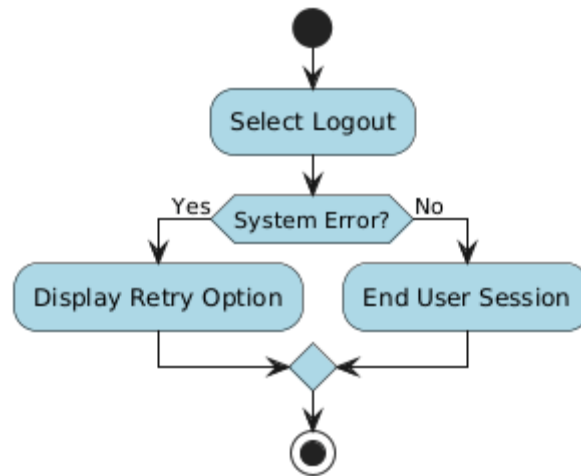


Figure 20: Activity Diagram for logout

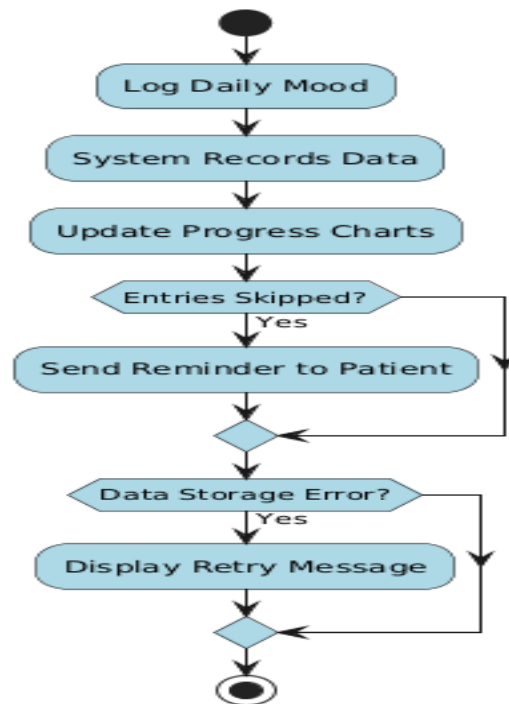


Figure 21: Activity Diagram for Mood and Progress Tracking

2.3.7 Sequence diagram

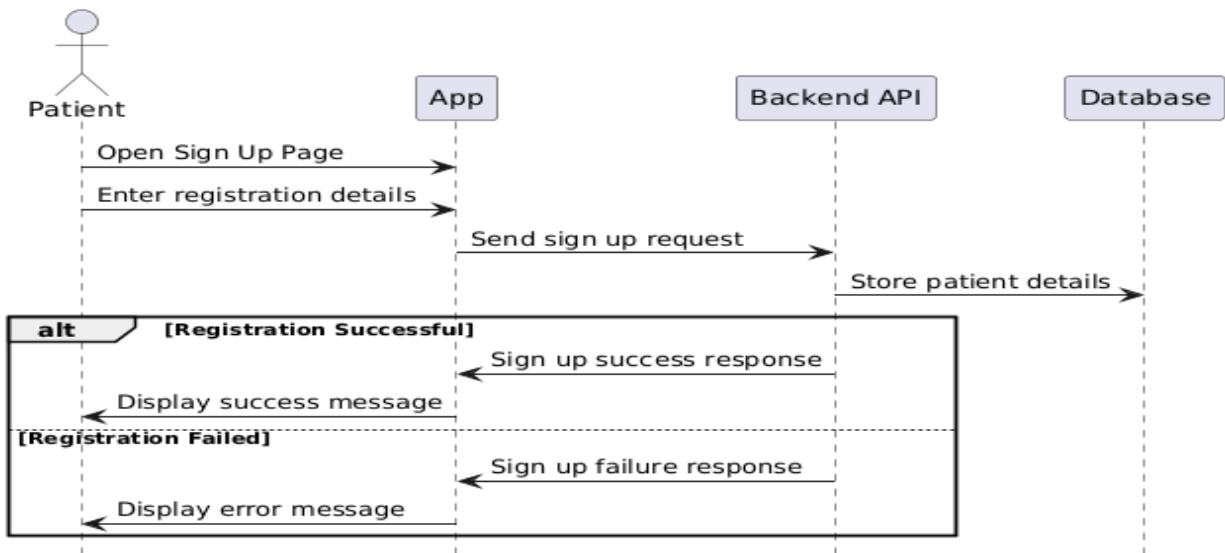


Figure 22: Sequence Diagram for sign up

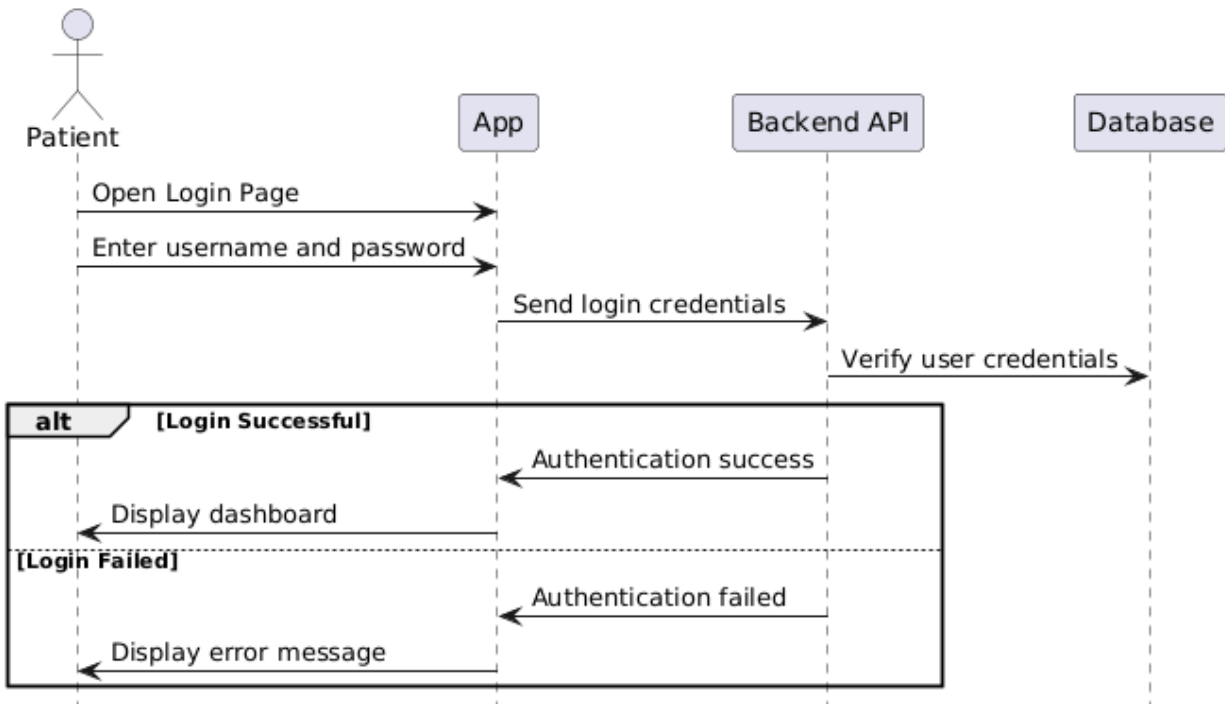


Figure 23: Sequence Diagram for login

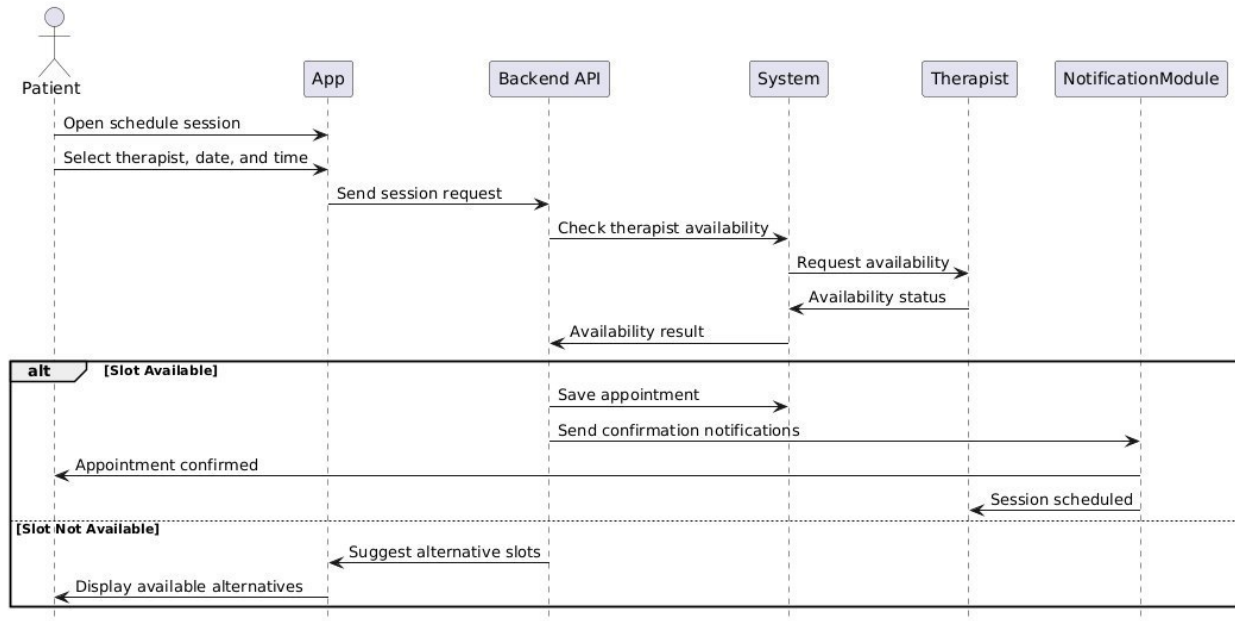


Figure 24: Sequence Diagram Therapy Session Scheduling

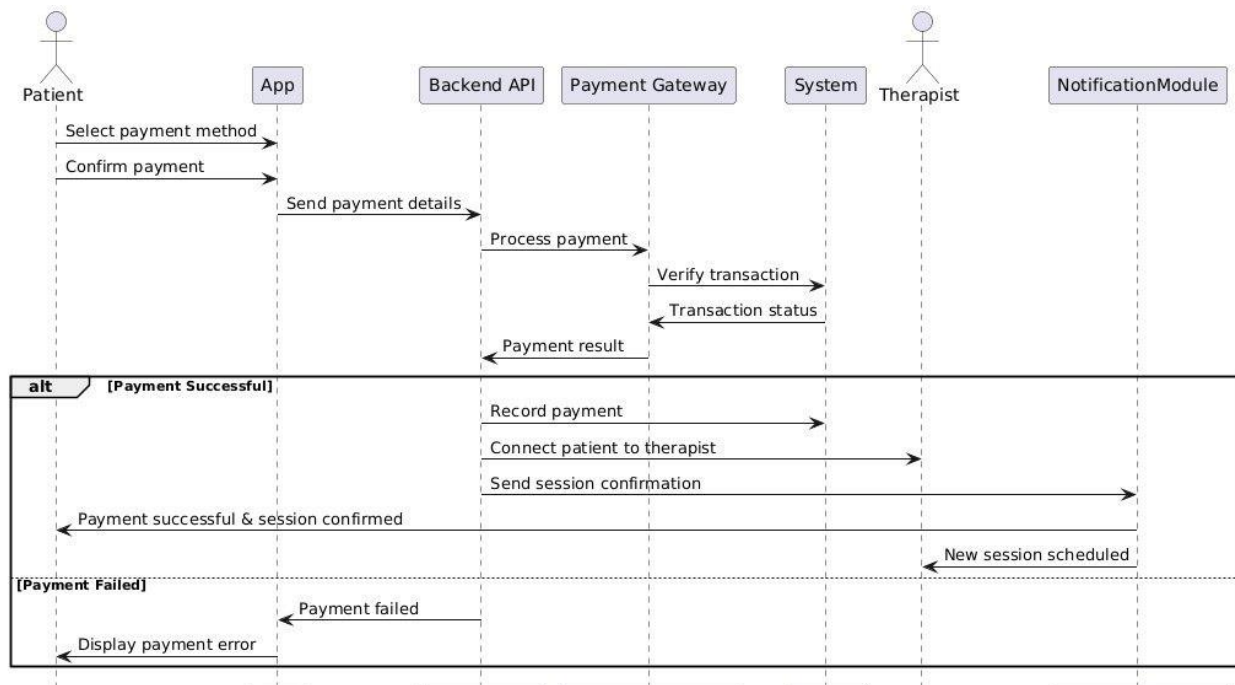


Figure 25: Sequence Diagram Payment for Payment

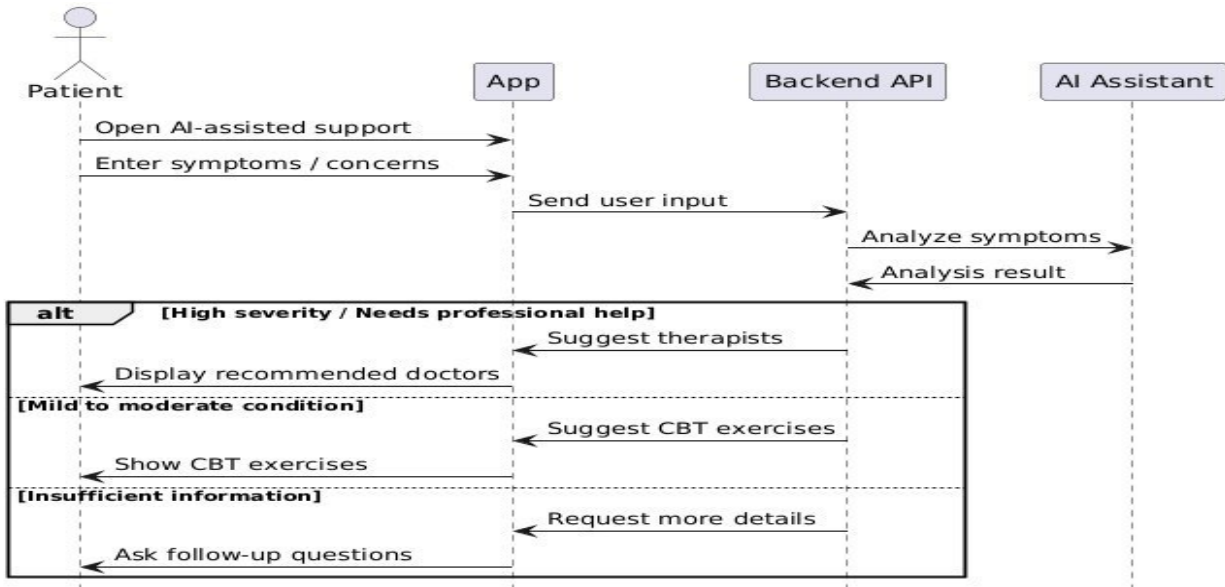


Figure 26: Sequence Diagram for AI Assistant

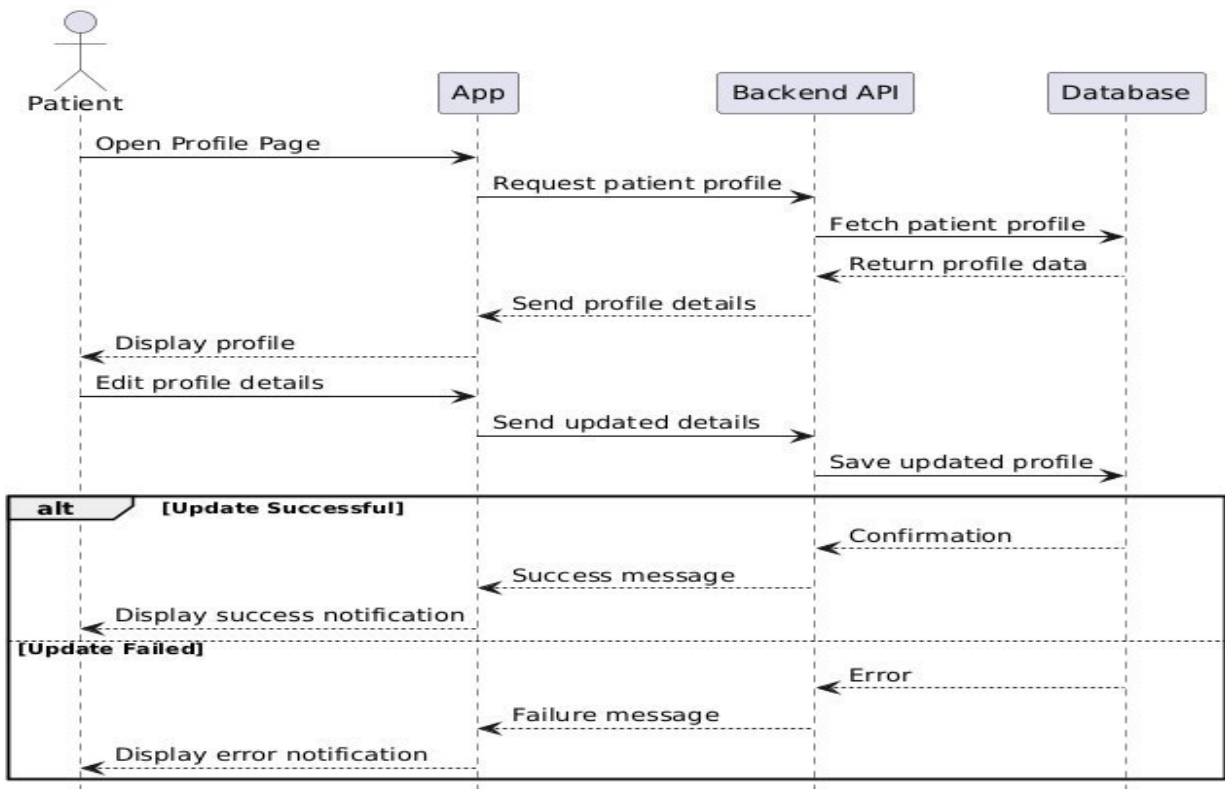


Figure 27: Sequence Diagram Profile Management

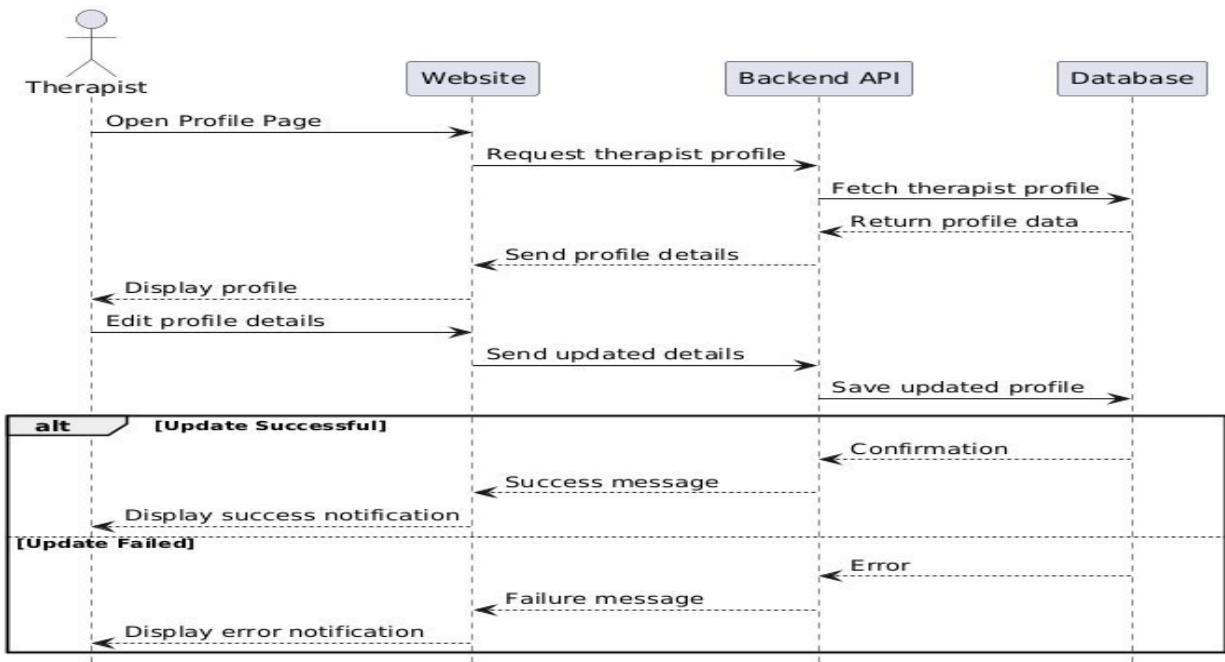


Figure 28: Sequence Diagram Therapist profile Management

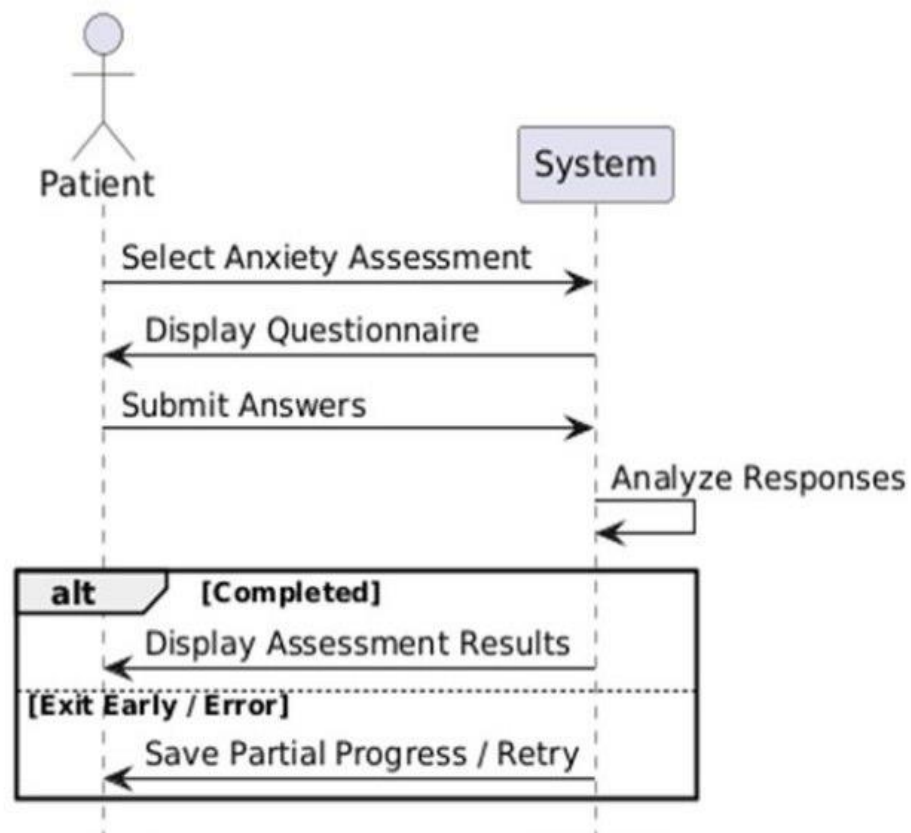


Figure 29: Sequence Diagram Anxiety Assessment

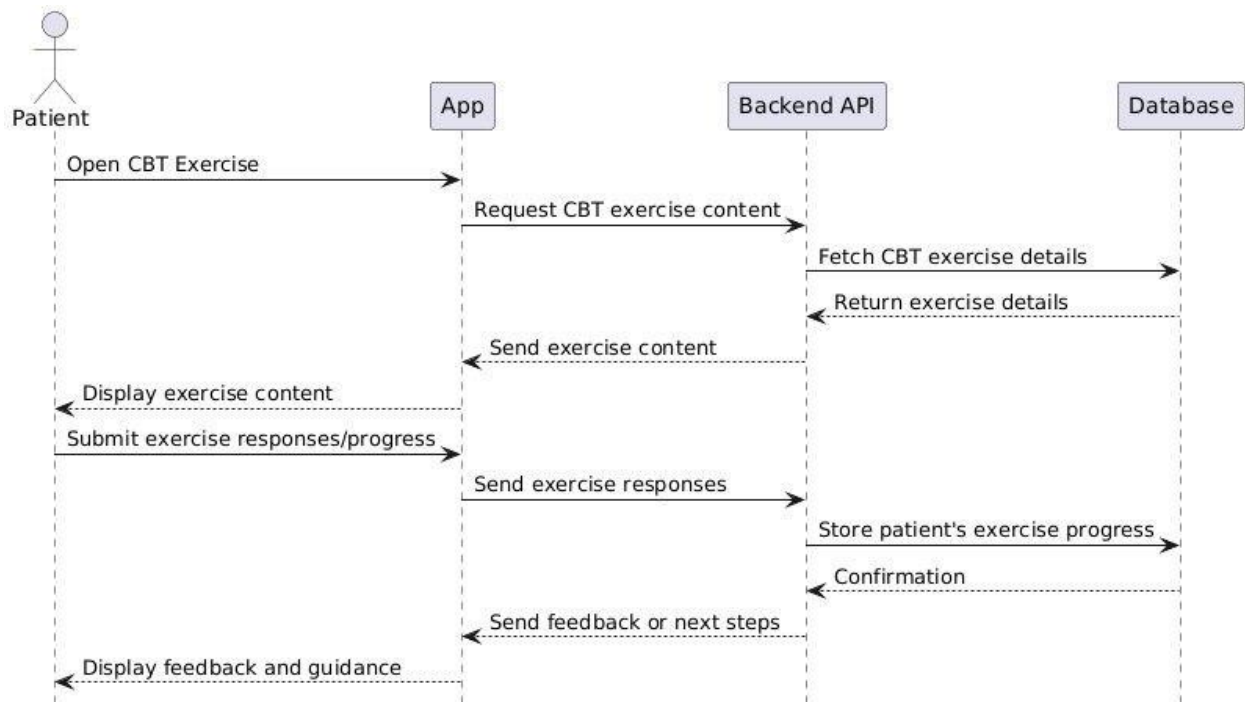


Figure 30: Sequence Diagram CBT Exercises

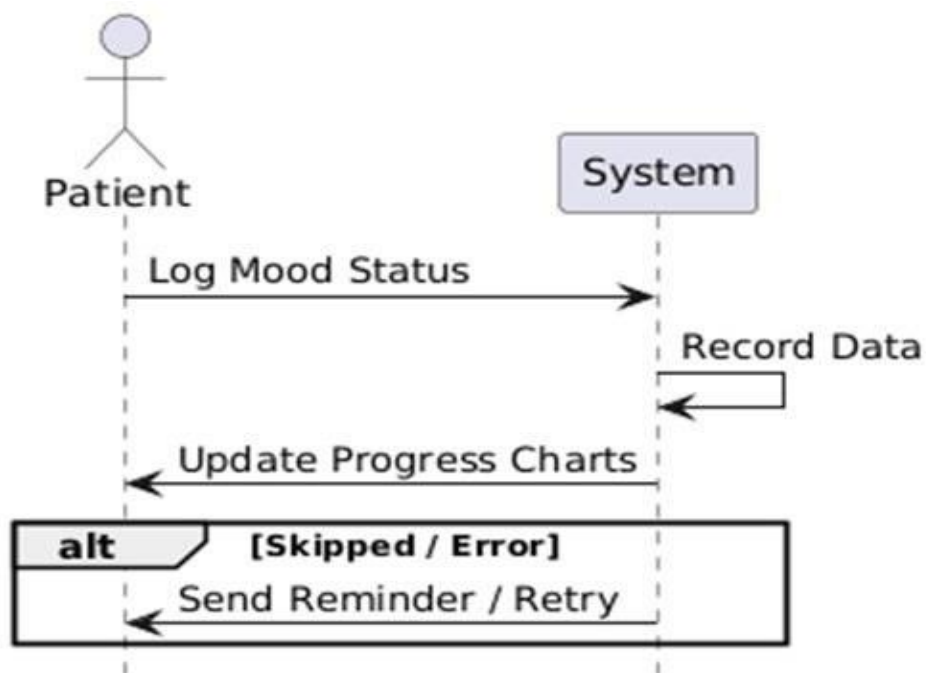


Figure 31: Sequence Diagram Mood and Progress Tracking

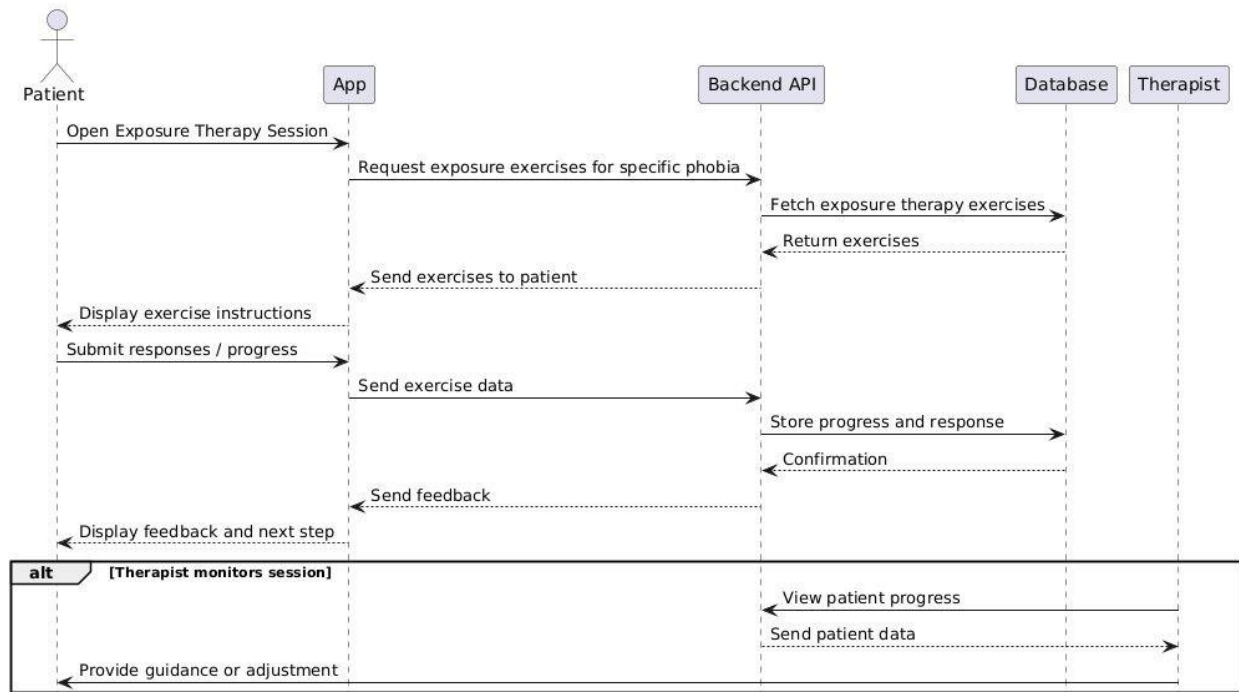


Figure 32: Sequence Diagram Exposure Therapy

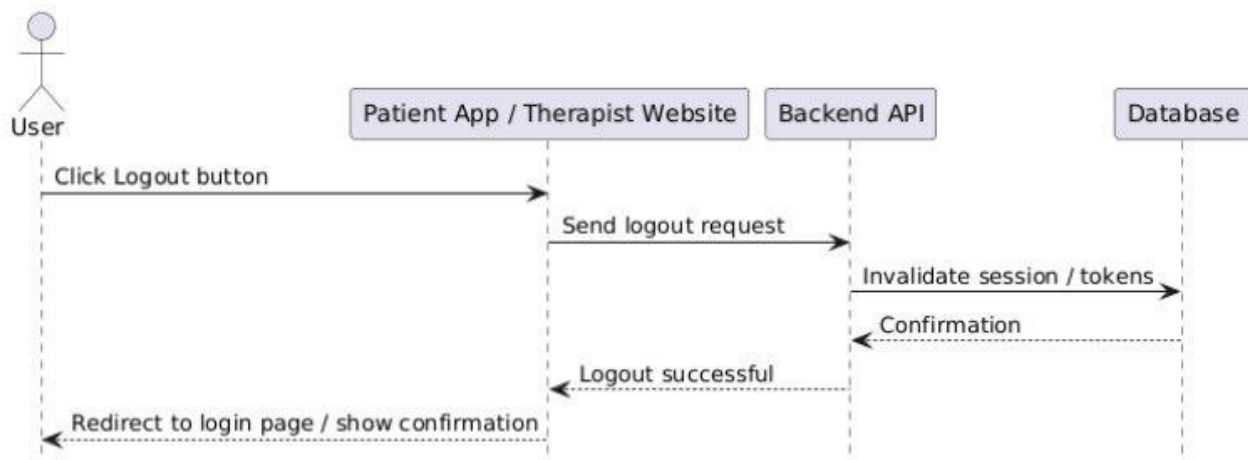


Figure 33: Sequence Diagram for logout

User

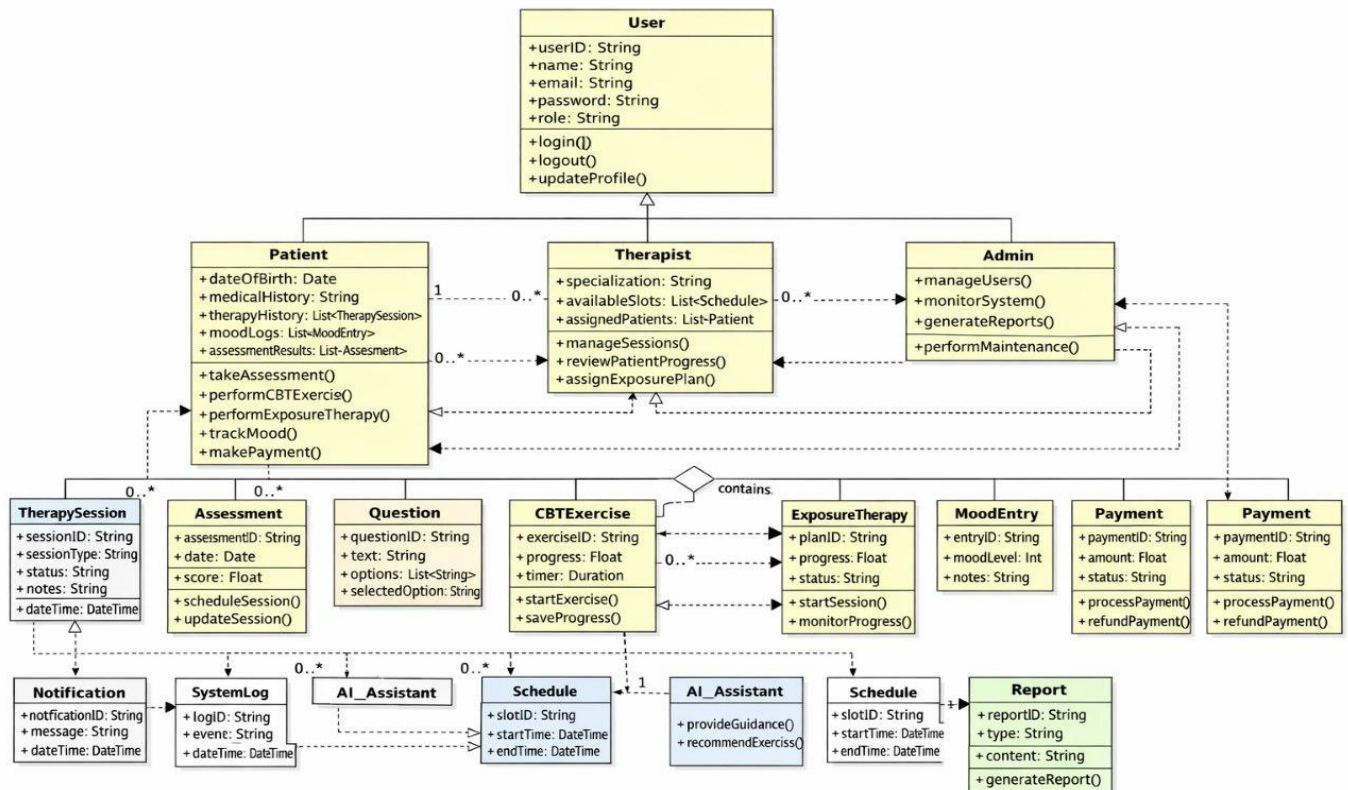


Figure 34: Analysis Class Model Diagram

2.3.9 Logic Model

Common logic model

START

- User Registration (Sign Up)

```
IF user selects "Sign Up" THEN
```

Display registration form

```
Input user details (name, email, password)
```

```
Validate input
```

IF valid THEN

Create user account

```
        Send confirmation email
        Display success message
    ELSE
        Display error message
        Request re-entry
    END IF
END IF
```

- **User Login**

```
IF user selects "Login" THEN
    Input credentials (email, password)
    Validate credentials
    IF valid THEN
        Redirect to user dashboard
    ELSE
        Display error
        Check max attempts
        IF max attempts exceeded THEN
            Lock account
        ELSE
            Retry login
        END IF
    END IF
END IF
```

- **Profile Management**

```
IF user selects "Edit Profile" THEN
    Input new profile data
```

```
    Validate changes
    Update profile
    Display confirmation
END IF
```

- **Anxiety Assessment**

```
IF user selects "Anxiety Assessment" THEN
    Display questionnaire
    Record responses
    Analyze results
    Display assessment results
END IF
```

- **Therapy Session Scheduling**

```
IF user selects "Schedule Session" THEN
    Display available therapists
    User selects therapist and preferred date/time
    Check therapist availability
    IF available THEN
        Confirm booking
        Send notification to patient and therapist
        Proceed to payment
    ELSE
        Suggest alternative slots
    END IF
END IF
```

- **Payment Processing**

```
IF session requires payment THEN
    Input payment details
    Process payment
    IF payment successful THEN
        Send receipt
        Update session status
    ELSE
        Display payment error
        Retry or cancel
    END IF
END IF
```

- **CBT / Exposure Therapy**

```
IF user selects CBT or Exposure Therapy THEN
    Display exercise module / exposure plan
    Monitor progress
    Save progress
    IF session interrupted THEN
        Save partial progress
    END IF
END IF
```

- **Mood & Progress Tracking**

```
IF user selects "Mood Tracking" THEN
    Input mood / progress data
    Record and save
    Update charts / reports
END IF
```

- **Emergency Support**

```
IF user clicks "Emergency Support" THEN
    Connect to emergency contacts
    Alert therapist/admin
    IF network failure THEN
        Display emergency numbers
    END IF
END IF
```

- **Logout**

```
IF user selects "Logout" THEN
    End user session
    Redirect to login page
END IF
END
```

2.4 Non-Functional Requirements

The non-functional requirements of the Therapy for Treating Phobias and Fear system ensure the platform operates efficiently, reliably, and securely while providing a seamless experience for patients, therapists, and administrators. These requirements focus on technical aspects such as performance, reliability, security, scalability, and usability.

1. Performance Requirements

The system must respond to user interactions (page loads, data fetches) within 2 seconds for 95% of operations.

2. Reliability Requirements

The system must achieve an uptime of 90%, ensuring minimal downtime for critical features such as therapy sessions, progress tracking, and emergency support.

The application must provide fallback mechanisms to ensure core functionalities like relaxation exercises and journal entries are accessible offline.

In case of system failures during therapy sessions, automatic reconnection mechanisms must be in place to resume sessions within 30 seconds/1min.

3. Security Requirements

All user data, especially Protected Health Information (PHI), must be encrypted in transit (TLS 1.2+) and at rest (AES-256). The system must implement multi-factor authentication for therapist and administrator accounts.

Role-Based Access Control (RBAC) must ensure that patients can only access their own data, and therapists can only access data for their assigned patients.

All text therapy sessions must be end-to-end encrypted to ensure patient confidentiality.

4. Usability Requirements

The app must provide a calming, intuitive interface designed to minimize user anxiety and stress. All interactive elements, especially within therapy modules, must provide clear instructions and feedback to the user. The system must support multiple languages to serve diverse patient populations. All content, including health insights and tips, must be localized to support Amharic, English, and other relevant languages.

5. Availability Requirements

The system must support 24/7 availability to ensure users can access therapy sessions, health insights, and other critical features at any time.

2.5 System Requirements

2.5.1 Hardware Requirements

Processor: Minimum 1 GHz; Recommended 2GHz or more

Hard Drive: Minimum 32 GB; Recommended 64 GB or more

Memory (RAM): Minimum 2 GB; Recommended 4 GB or above

2.5.2 Software Requirements

Browser Requirements:

The system supports major modern web browsers. Users can choose the browser that works best on their devices. Recommended browsers include: Firefox ,Chrome, Opera and Microsoft Edge.

Operating System:Windows 10 or later (including Windows 11)

Development Tools:

- **IDE:** Android Studio (latest version), Visual Studio Code with Dart and Flutter extensions
- **Programming Languages:** Dart (using the Flutter framework for frontend and mobile applications) , Node.js (for backend development, chatbot integration, and data processing)
- **Version Control:** Git and GitHub for collaboration and repository management

Database:

- **Primary Database:** Firebase Realtime Database or Firestore for real-time data synchronization and efficient mobile integration
- **Backup Database:** PostgreSQL for structured data storage and analytics

Security Tools:

- JWT (JSON Web Tokens) for secure API and web communication
- Encryption tools for protecting sensitive data

Testing Tools:

- Flutter's built-in testing frameworks for unit, integration, and widget testing
- Postman for API testing
- Selenium for web-based functionality testing

2.6 Key Abstraction with CRC Analysis

2.6.1 Conceptual Modeling: Class Diagram

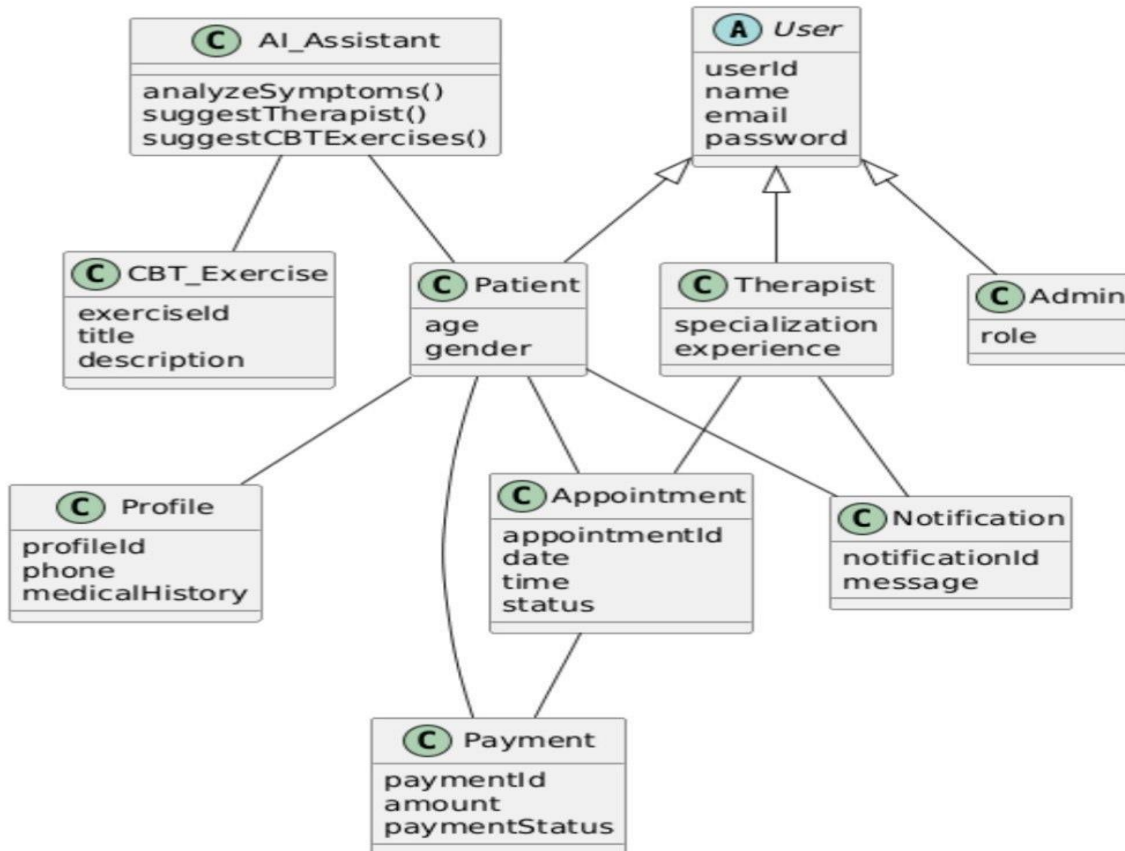


Figure 35: Conceptual Model Class Diagram

2.6.2 Identifying change cases

Change Case	Description	Likelihood	Impact
Add a new user role	Introduce a role, such as user, to assist admins.	Moderate	Modify class diagram and access rules.
Extend chatbot functionality	Enable the chatbot to provide emotional support or integrate machine learning.	High	Add methods and attributes to the Chatbot class.
Modify health insights	Include content like messages.	High	Update HealthInsight attributes.
Introduce recurring appointments	Allow users to schedule recurring health checkups.	Moderate	Modify the Appointment class to include recurrence logic.
Enhance chat	Add media sharing in chat messages.	High	Extend the message Chatclass

Table 16: change cases description

2.6.3 User Interface Prototyping

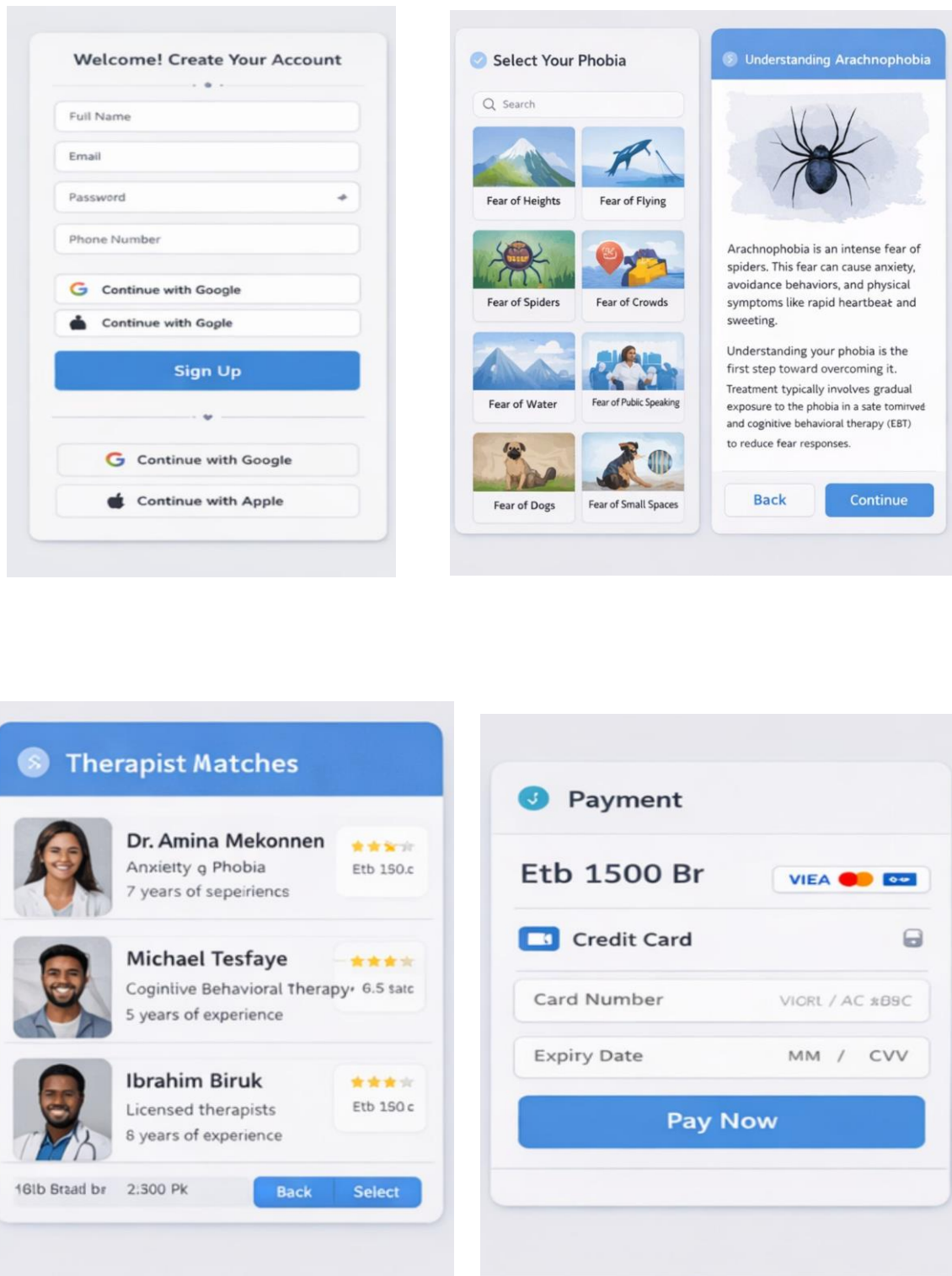


Figure 36: patient UI



Figure 37 :System Administrator UI

Chapter Three: System Design

This chapter presents the system design of the Anxiety and Phobia Treatment System, transforming the requirements defined in the previous chapter into a structured technical blueprint. The design phase focuses on how the system will be organized, how components interact, and how quality attributes such as performance, dependability, and usability are achieved.

The system is designed as a digital mental health support platform that assists users in managing anxiety and phobia through self-assessment tools, guided therapy modules, professional support, and progress tracking. Emphasis is placed on privacy, reliability, and user comfort to ensure a safe and supportive environment for end users.

The design decisions described in this chapter guide developers in implementing a system that is scalable, secure, and user-centered.

Design Goals

Design goals define the quality attributes the system must optimize to satisfy non-functional requirements. For the Anxiety and Phobia Treatment System, the key design perspectives include functionality, performance, dependability, maintainability, and end-user experience. These goals ensure the system is effective, reliable, and easy to use for individuals seeking mental health support.

Functionality

The system is designed to fulfill all functional requirements related to anxiety and phobia management. Core functional goals include:

- Providing users with self-assessment questionnaires to identify anxiety or phobia levels.
- Offering guided therapy programs such as relaxation exercises, cognitive behavioral techniques, and exposure guidance.
- Enabling secure communication between users and mental health professionals (if applicable).
- Allowing users to track progress, view reports, and receive personalized recommendations.
- Supporting user account management, including registration, login, and profile customization.

The focus on functionality ensures the system addresses the diverse needs of users with varying levels of anxiety or phobia.

Performance

Performance is critical for a mental health system, as delays or interruptions may negatively affect user trust and engagement. The system is designed to provide:

- Fast response times for assessments, content loading, and progress updates.
- Efficient handling of multiple concurrent users, especially during peak usage hours.
- Smooth delivery of multimedia content such as audio relaxation guides and instructional videos.
- Optimized database queries to ensure quick retrieval of user data and therapy content.

The system aims to maintain consistent performance across different devices and network conditions.

Dependability

Dependability ensures that users can rely on the system at all times, particularly when they are in need of emotional support.

1. Reliability

The system is designed to function correctly under normal conditions with minimal failures. Reliable cloud hosting, validated inputs, and stable backend services ensure continuous operation.

2. Availability

The platform targets high availability, allowing users to access therapy resources and self-help tools 24/7, regardless of location.

3. Security

Security is a top priority due to the sensitive nature of mental health data. The system implements:

- Secure authentication and authorization mechanisms.
- Encryption of sensitive user data.
- Controlled access to personal records and therapy progress.

- Secure data storage and backup mechanisms.

4. Error Handling

The system includes robust error-handling strategies:

- Input validation to prevent incorrect or harmful data entry.
- Exception handling and logging to detect and diagnose system errors.
- User-friendly error messages that avoid technical jargon and reduce stress.

Maintainability

Maintainability ensures the system can be updated, improved, and fixed efficiently over time.

1. Documentation

Clear and structured documentation is maintained for system architecture, APIs, and user workflows to support future development and maintenance.

2. Code Readability

Modular design, consistent coding standards, and meaningful naming conventions improve readability and reduce maintenance complexity.

3. Monitoring and Logging

System logs and monitoring tools help identify errors early and ensure stable operation.

4. Scalability

The system architecture supports future expansion, such as adding new therapy modules, increasing the number of users, or integrating wearable health data.

End-User Criteria

End-user considerations are central to the system design, as users may already be experiencing stress or anxiety.

1. User-Friendly Interface

The interface is simple, calming, and intuitive, using clear navigation and minimal visual clutter.

2. Comfort and Accessibility

The system uses soothing colors, readable fonts, and accessible design to reduce cognitive load and emotional discomfort.

3. Personalization

Users receive personalized therapy suggestions based on their assessment results and usage history.

4. Privacy and Trust

Transparent privacy policies, secure data handling, and clear consent mechanisms help build user confidence.

5. Convenience

Users can access therapy resources anytime without needing physical visits, making mental health support more accessible and flexible.

3.1 Architectural Design

The Anxiety and Phobia Treatment System is designed as a web-based application that follows a three-tier architecture model. This architectural approach improves scalability, maintainability, security, and performance by separating the system into clearly defined layers, each responsible for a specific set of functionalities.

The architecture supports users seeking mental health assistance through self-assessment tools, therapy modules, and progress tracking, while ensuring data privacy and system reliability.

The system is organized into the following three layers:

Three-Tier Architecture Overview

1. Presentation Layer (Client-Side UI)

The Presentation Layer represents the user-facing part of the system. It enables interaction between users and the system through a mobile application.

Responsibilities:

- Displaying user interfaces for login, registration, and profile management
- Presenting anxiety and phobia self-assessment questionnaires
- Displaying therapy content such as relaxation exercises and educational materials
- Showing progress reports and feedback dashboards

Communication:

- This layer communicates with the Application Layer using HTTP/HTTPS requests
- It does not directly access the database, ensuring strong separation of concerns

2. Application Layer (Business Logic & API Server)

The Application Layer serves as the core processing unit of the system. It handles all business logic and coordinates interactions between the Presentation and Data layers.

Responsibilities:

- Processing user requests and managing therapy workflows
- Evaluating assessment results and generating personalized recommendations
- Managing user authentication and authorization
- Enforcing security policies and validating user input
- Coordinating data exchange with the Data Layer

Role in Architecture:

- Acts as middleware between the UI and database
- Ensures consistency, integrity, and security of system operations

3. Data Layer (Database Server & Storage Services)

The Data Layer is responsible for data storage, retrieval, and management. Due to the sensitive nature of mental health information, this layer is designed with strong data protection mechanisms.

Responsibilities:

- Storing user profiles and credentials
- Storing assessment results and therapy progress data
- Managing therapy content and system configuration data
- Supporting backup and recovery mechanisms

Access Control:

- Only the Application Layer can interact with the Data Layer
- This restriction improves security and data integrity

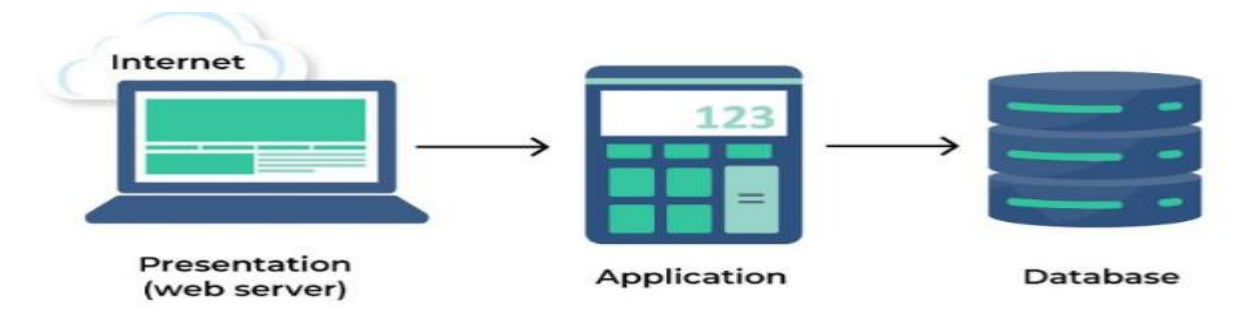


Figure 38: Three-tier architecture model

Subsystem Design

System decomposition is a crucial aspect of software architecture that divides a complex system into smaller, manageable subsystems. Each subsystem is a self-contained unit consisting of related components, operations, and constraints that collectively perform a specific function.

The subsystems of the Anxiety and Phobia Treatment System are designed to work together while maintaining high cohesion and low coupling, ensuring flexibility and ease of maintenance.

Identified Subsystems

1. User Interface Subsystem

Services Provided:

- User interaction and navigation
- Display of assessments, therapy modules, and progress reports

Services Accepted:

- Authentication responses
- Therapy recommendations and user data

This subsystem focuses solely on presentation logic, ensuring high cohesion.

2. User Management Subsystem

- User registration and authentication
- Profile creation and updates

Services Accepted:

- Login and registration requests from the UI

This subsystem ensures secure access control and identity management.

3. Assessment and Therapy Subsystem

Services Provided:

- Anxiety and phobia assessments
- Therapy plans and guided exercises
- Personalized treatment recommendations

Services Accepted:

- User responses from the UI
- Historical data from the Data Management Subsystem

4. Progress Monitoring Subsystem

Services Provided:

- Progress tracking and performance analysis
- Generation of reports and visual feedback

Services Accepted:

- Assessment results and therapy completion data

5. Notification and Reminder Subsystem

Services Provided:

- Therapy reminders and motivational notifications

Services Accepted:

- Scheduling data from the Application Layer

6. Data Management Subsystem

Services Provided:

- Data storage and retrieval services
- Secure handling of sensitive mental health records

Services Accepted:

- Data requests from all other subsystems via the Application Layer

Subsystem Interaction

- The User Interface Subsystem communicates with the User Management and Assessment and Therapy Subsystems
- The Assessment and Therapy Subsystem interacts with the Progress Monitoring Subsystem
- All subsystems rely on the Data Management Subsystem for persistent storage
- Communication occurs through well-defined APIs, ensuring loose coupling

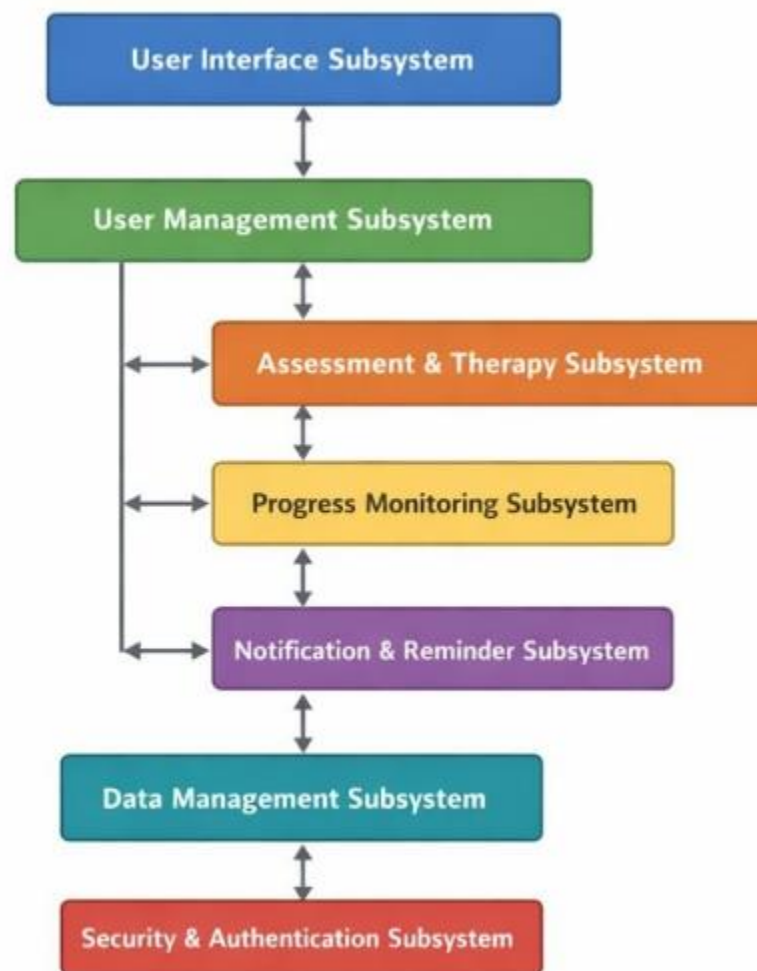


Figure 39: Subsystem Model for Anxiety and phobia Treatment System

3.1.1 Component Modeling

In a component-based architecture, the Anxiety and Phobia Treatment System is divided into modular, self-contained units called components. Each component is responsible for a specific set of functionalities or services, contributing to the overall system in a modular, maintainable, and scalable way.

Components are loosely coupled, allowing them to operate independently while collaborating through well-defined interfaces. This architecture ensures flexibility, maintainability, and scalability, and enables the system to handle multiple operations such as user management, therapy sessions, progress monitoring, notifications, and data management without a single component becoming a bottleneck.

Component interactions occur via APIs or service calls, allowing seamless communication and data transfer. Each component focuses on its specific responsibility but can request or provide services to other components as needed.

User Interface Component

The User Interface (UI) Component serves as the primary interaction point for users of the system, including patients, therapists, and administrators. This component allows users to access services and interact with the system efficiently through a web or mobile application.

The UI component interacts with the following components:

- **User Management Component Interface:** Enables users to register, login, update profiles, and manage authentication securely.
- **Assessment & Therapy Component Interface:** Allows patients to participate in therapy sessions, complete assessments, and access personalized treatment plans.
- **Progress Monitoring Component Interface:** Displays patient progress, therapy history, and analytics dashboards for both patients and therapists.
- **Notification & Reminder Component Interface:** Manages reminders for therapy sessions, assessment deadlines, and motivational notifications.
- **Data Management Component Interface:** Ensures secure retrieval and storage of user data, session records, assessments, and progress reports.
- **Security & Authentication Component Interface:** Provides secure authentication, encryption of sensitive data, and role-based access control.

User Management Component

This component handles all aspects of user authentication, registration, and role management. It manages different user roles, such as patients, therapists, and administrators, and ensures secure access to system features.

- Interfaces with UI Component for user interaction.
- Interfaces with Data Management Component to store and retrieve user profiles and authentication data.
- Interfaces with Security & Authentication Component to validate user credentials and manage sessions.

Assessment & Therapy Component

This component is responsible for managing therapy sessions and assessments for anxiety and phobia patients. It provides:

- Personalized therapy programs.
- Interactive self-help exercises and guided sessions.
- Assessment tools to evaluate anxiety and phobia levels.
- Interfaces with UI Component to display therapy content.
- Interfaces with User Management Component to access patient profiles.
- Interfaces with Progress Monitoring Component to track therapy progress and update analytics.

Progress Monitoring Component

This component tracks and reports on patient treatment progress, session history, and therapy effectiveness. It generates visual reports and analytics dashboards for patients and therapists.

- Interfaces with Assessment & Therapy Component for session data.
- Interfaces with UI Component for presenting progress insights.
- Interfaces with Data Management Component to store monitoring data.

Notification & Reminder Component

This component manages all system-generated notifications and reminders, including:

- Therapy session reminders.
- Assessment completion notifications.
- Motivational and alert messages to encourage user engagement.
- Interfaces with UI Component to display notifications.
- Interfaces with Data Management Component to schedule and store reminders.

Data Management Component

This component is responsible for data storage, retrieval, and management. It ensures the secure handling of sensitive user and therapy data. It also provides interfaces for other components to access data reliably and efficiently.

- Interfaces with all other components for data storage and retrieval.
- Maintains a secure and structured database of users, therapy content, assessments, and progress reports.

Security & Authentication Component

This component ensures system security, data privacy, and access control. It handles:

- User authentication and session management.
- Role-based access control.
- Data encryption and secure communication between components.
- Interfaces with User Management Component for authentication.
- Interfaces with UI Component to enforce secure access.
- Interfaces with Data Management Component to protect stored data.

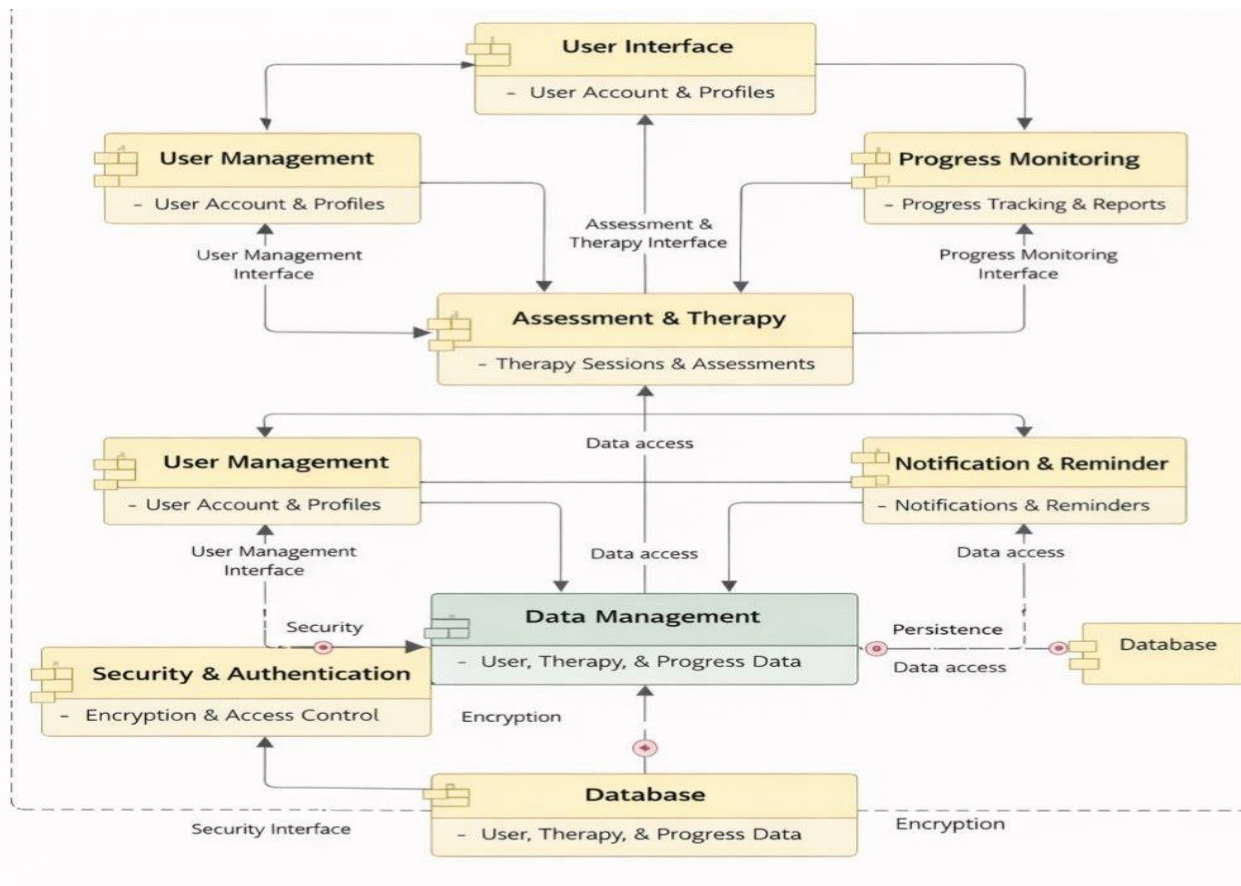


Figure 40: Component Diagram for Anxiety and phobia Treatment System

3.1.2 Deployment Modeling

The Anxiety and Phobia Treatment System is deployed using a three-tier architecture model, distributing software components across multiple servers and devices. This approach ensures optimal performance, maintainability, security, and scalability. Each tier is assigned to dedicated infrastructure to ensure efficient resource utilization, reliable communication, and secure data handling.

1. Client Layer (Presentation Layer Deployment)

Client Devices (Mobile)

- Users interact with the system through mobile applications.
- All user actions, such as scheduling therapy sessions, completing assessments, or receiving notifications, are sent to the Application Server via HTTP/HTTPS protocols.
- Technologies used: React Native for mobile apps, React.js for web frontend, HTML, CSS, and JavaScript.

Hosting Environment

- The frontend is deployed on a cloud-based Content Delivery Network (CDN) to enhance performance and minimize latency.
- Users experience fast loading times and high availability regardless of geographic location.

2. Application Layer (Business Logic & API Server Deployment)

Web & Application Server

- Hosts both frontend and backend services, processing business logic, authentication, and authorization.
- Handles core functionalities such as:
 - **Assessment & Therapy Sessions:** Managing mental health assessments, therapy exercises, and session scheduling.
 - **Progress Monitoring:** Tracking user progress and generating reports.
 - **Notification & Reminders:** Sending alerts for upcoming sessions, medication schedules, or therapy tasks.
- Manages communication with external services such as payment gateways for premium features or subscription services.

Security Features

- API authentication and authorization protocols (JWT or OAuth2).
- Encryption for sensitive data (e.g., user therapy records).
- Rate limiting and firewall protection to prevent abuse and attacks.

Hosting Environment

- Deployed on cloud-based servers (e.g., AWS EC2, Azure, or Google Cloud) to ensure scalability, load balancing, and high availability.

3. Data Layer (Database Server & External Integrations Deployment)

Database Server

- Uses PostgreSQL to store user profiles, therapy session data, assessment results, progress reports, and notification history.
- Data management is implemented with an ORM (Object-Relational Mapping) for efficient database interaction and consistency.

External Integrations

- Secure communication with payment services for handling premium subscriptions or therapy packages.
- Optional integration with mental health APIs for guided exercises or AI-based therapy recommendations.

Communication Flow

1. **User Request:** A user initiates an action on the Client Device (web or mobile).
2. **Frontend Processing:** The request is forwarded to the Application Server, where business logic validates the request.
3. **Data Operations:**
 - For data retrieval or storage, the Application Server communicates with the Database Server.
 - For subscription payments, the Application Server interacts with the Payment Gateway API.
4. **Response Delivery:** Processed results are sent back through the Application Server to the Client Device, ensuring real-time updates and notifications.

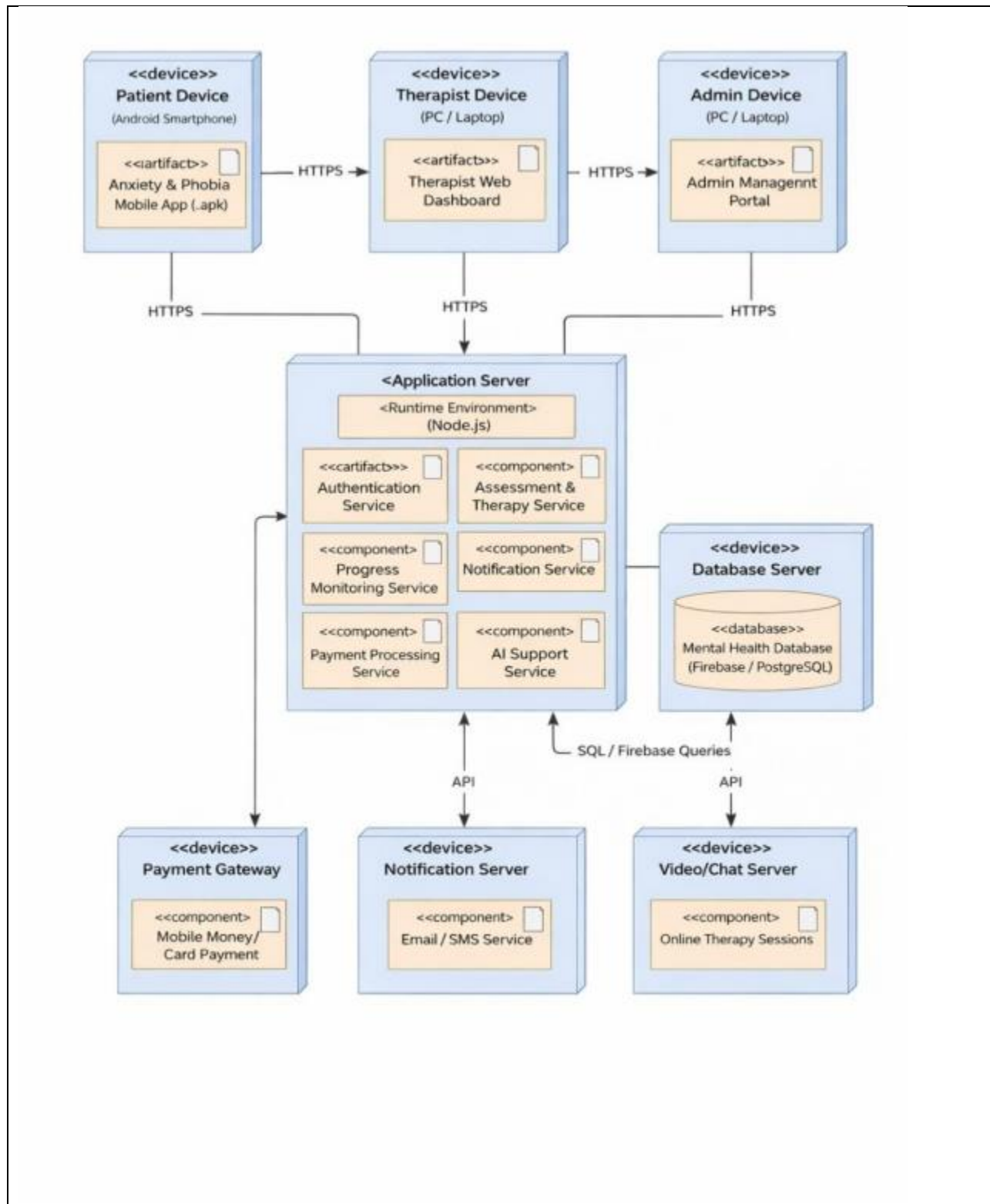


Figure 41: Deployment Diagram for Anxiety and Phobia Treatment System

Justification for the Architecture

The deployment design of the Anxiety and Phobia Treatment System uses a multi-tier architecture for the following advantages:

1. **Modularity:** Each layer (Presentation, Application, Data) is independent, making it easier to maintain, upgrade, or replace without affecting other layers.
2. **Scalability:** Each component can be scaled individually, e.g., more application servers can be added to handle high user loads without altering the database or frontend.
3. **Security:** Direct access to the database is restricted; all client interactions pass through the Application Server. Sensitive user data such as therapy records are encrypted.
4. **Performance Optimization:** By distributing components across multiple servers and using cloud infrastructure, system bottlenecks are minimized, ensuring fast response times for users.
5. **Reliability:** Cloud deployment with load balancing and redundant servers ensures high availability and minimal downtime for users accessing therapy sessions or assessments.

3.2 Detail Design

3.2.1 Design class model

The Design Class Model represents the object-oriented structure of the Therapy for treating phobias and Anxiety system, defining classes, attributes, methods, relationships, access visibility, and multiplicity. This model ensures a clear understanding of how different components interact within the system.

Each class in the system is described with:

Attributes: Data members, their data types, and access modifiers (public, private, protected).

Methods: Functions associated with the class, specifying return types and parameters with data types.

Relationships: Associations between classes (one-to-one, one-to-many, etc.), including inheritance and dependencies.

Multiplicity: Defines how many instances of one class relate to another (e.g., 1, 0..*, 1..*).

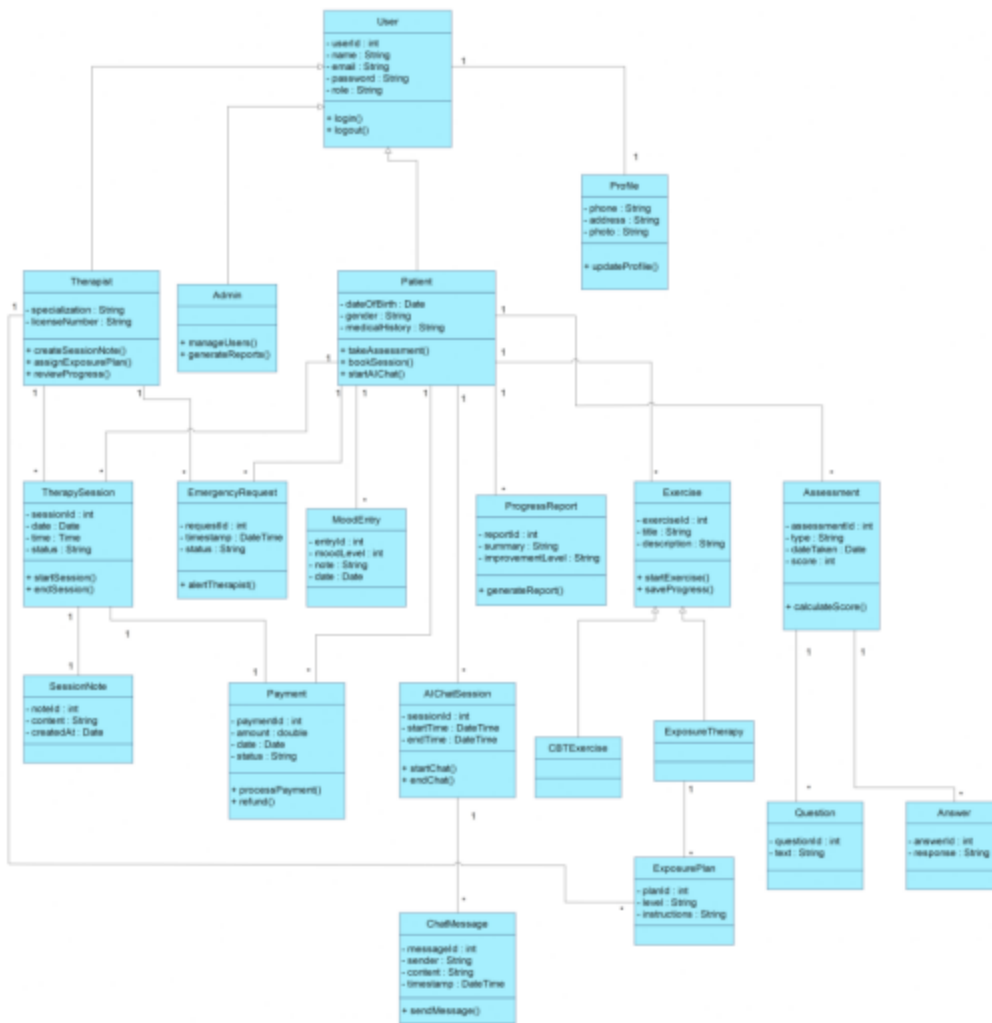


Figure 42:Class Model

Database Model – Entity Tables

1. Roles

Field	Type	Notes
role_id	INT (PK)	Unique identifier for role
role_name	VARCHAR(50)	Name of the role
description	TEXT	Role description

2. Users

Field	Type	Notes
user_id	INT (PK)	Unique user identifier
email	VARCHAR(255)	User email (unique)
password_hash	VARCHAR(255)	Encrypted password
first_name	VARCHAR(100)	User first name
last_name	VARCHAR(100)	User last name
role_id	INT (FK)	References Roles
phone	VARCHAR(20)	Contact number
is_active	BOOLEAN	Account status
created_at	TIMESTAMP	Account creation date

3. UserConsents (Ethical Requirement)

Field	Type	Notes
consent_id	INT (PK)	Consent record ID
user_id	INT (FK)	References Users
consent_type	VARCHAR(255)	Privacy / Terms / Data usage

accepted	BOOLEAN	Consent status
accepted_at	TIMESTAMP	Acceptance date

4. LoginAuditLogs

Field Name	Data Type	Description
log_id	INT (PK)	Log identifier
user_id	INT (FK)	References Users
login_time	TIMESTAMP	Login time
logout_time	TIMESTAMP	Logout time
ip_address	VARCHAR(50)	User IP
status	VARCHAR(20)	Success / Failed

5. Patients

Field	Type	Notes
patient_id	INT (PK/FK)	References Users
date_of_birth	DATE	Date of birth
emergency_contact_name	VARCHAR(255)	Emergency contact
emergency_contact_phone	VARCHAR(20)	Emergency phone
primary_phobia	VARCHAR(100)	Main phobia
secondary_phobias	TEXT	Additional phobias
current_anxiety_level	INT	Anxiety level (1–10)
notes	TEXT	Clinical notes

6. Therapists

Field	Type	Notes
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therapist_id	INT (PK/FK)	References Users
license_number	VARCHAR(100)	Professional license
specialization	VARCHAR(255)	Area of expertise
years_of_experience	INT	Experience in years
hourly_rate	DECIMAL(10,2)	Session rate
is_verified	BOOLEAN	Verification status

7. Admins

Field	Type	Notes
admin_id	INT (PK/FK)	References Users
access_level	ENUM	Administrative access level
created_at	TIMESTAMP	Account creation time

8. Therapist Patient Assignments

Field	Type	Notes
assignment_id	INT (PK)	Assignment identifier
patient_id	INT (FK)	References Patients
therapist_id	INT (FK)	References Therapists
start_date	DATE	Assignment start
end_date	DATE	Assignment end
is_active	BOOLEAN	Current assignment status

9. Appointments

Field	Type	Notes
appointment_id	INT (PK)	Session identifier
patient_id	INT (FK)	References Patients

therapist_id	INT (FK)	References Therapists
appointment_date	DATE	Session date
start_time	TIME	Start time
end_time	TIME	End time
status	VARCHAR(50)	Session status
session_type	VARCHAR(50)	CBT / Exposure

10.SessionNotes

Field Name	Data Type	Description
note_id	INT (PK)	Note ID
appointment_id	INT (FK)	References Appointments
therapist_id	INT (FK)	References Therapists
patient_id	INT (FK)	References Patients
subjective	TEXT	Patient report
objective	TEXT	Therapist observation
assessment	TEXT	Clinical assessment
plan	TEXT	Treatment plan
created_at	TIMESTAMP	Date recorded

11.AnxietyAssessments

Field	Type	Notes
assessment_id	INT (PK)	Assessment identifier
patient_id	INT (FK)	References Patients
therapist_id	INT (FK)	References Therapists

assessment_date	DATE	Assessment date
total_score	INT	Calculated anxiety score
severity_level	ENUM	Severity classification

11.AssessmentQuestions

Field	Type	Notes
question_id	INT (PK)	Question identifier
question_text	TEXT	Assessment question
category	VARCHAR(50)	Anxiety / Phobia

12.AssessmentResponses

Field	Type	Notes
response_id	INT (PK)	Response identifier
assessment_id	INT (FK)	References AnxietyAssessments
question_id	INT (FK)	References AssessmentQuestions
response_value	INT	User response score

13.CBTExercises

Field	Type	Notes
exercise_id	INT (PK)	Exercise identifier
title	VARCHAR(255)	Exercise title
description	TEXT	Exercise description
difficulty_level	VARCHAR(50)	Easy / Medium / Hard

14.CBT Progress

Field	Type	Notes
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progress_id	INT (PK)	Progress identifier
patient_id	INT (FK)	References Patients
exercise_id	INT (FK)	References CBTEercises
status	ENUM	Completion status
feedback	TEXT	Patient feedback
completed_at	TIMESTAMP	Completion date

15.ExposureTherapySessions

Field	Type	Notes
session_id	INT (PK)	Exposure session identifier
patient_id	INT (FK)	References Patients
therapist_id	INT (FK)	References Therapists
exposure_level	INT	Exposure intensity (1–10)
anxiety_start	INT	Anxiety at start
anxiety_end	INT	Anxiety at end
notes	TEXT	Session notes
session_date	DATE	Session date

16.MoodTracking

Field	Type	Notes
track_id	INT (PK)	Mood tracking identifier
patient_id	INT (FK)	References Patients
tracking_date	DATE	Date of entry
mood_level	INT	Mood score
anxiety_level	INT	Anxiety score
notes	TEXT	Additional notes

17.AI_Interactions

Field Name	Data Type	Description
interaction_id	INT (PK)	Interaction ID
patient_id	INT (FK)	References Patients
user_message	TEXT	Patient input
ai_response	TEXT	AI reply
created_at	TIMESTAMP	Interaction time

18.Payments

Field	Type	Notes
payment_id	INT (PK)	Payment identifier
appointment_id	INT (FK)	Related appointment
patient_id	INT (FK)	Paying patient
amount	DECIMAL(10,2)	Payment amount
payment_method	VARCHAR(50)	Payment method
transaction_id	VARCHAR(100)	Gateway reference
status	ENUM	Payment status
payment_date	TIMESTAMP	Payment time

18.EmergencySupportLogs

Field	Type	Notes
log_id	INT (PK)	Emergency log identifier
patient_id	INT (FK)	References Patients
triggered_at	TIMESTAMP	Alert time
anxiety_level	INT	Reported anxiety

resolution	TEXT	Resolution details
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19.SessionNotes

Field	Type	Notes
note_id	INT (PK)	Note identifier
appointment_id	INT (FK)	References Appointments
therapist_id	INT (FK)	References Therapists
patient_id	INT (FK)	References Patients
subjective	TEXT	Patient-reported
objective	TEXT	Therapist observation
assessment	TEXT	Clinical assessment
plan	TEXT	Treatment plan

20.Notifications

Field	Type	Notes
notification_id	INT (PK)	Notification identifier
user_id	INT (FK)	References Users
title	VARCHAR(255)	Notification title
message	TEXT	Notification message
is_read	BOOLEAN	Read status
created_at	TIMESTAMP	Time sent

21.SystemLogs

Field	Type	Notes
log_id	INT (PK)	Log identifier
user_id	INT (FK)	References Users

action	VARCHAR(100)	Performed action
created_at	TIMESTAMP	Log timestamp

Table 17: Database Entity Table

3.2.2 Entity-Relationship (E-R) Model

The Entity-Relationship (E-R) Model represents the logical structure of the Therapy for treating phobias and Anxiety relational database. It defines how data is organized, stored, and related within the system.

The E-R Model consists of the following key elements:

- **Entities:** Represent real-world objects.
- **Attributes:** Define the properties of each entity, including data types and constraints.
- **Primary Keys (PKs):** Unique identifiers for each entity.
- **Foreign Keys (FKs):** Establish relationships between different entities.
- **Relationships:** Define how entities are connected (one-to-one, one-to-many, many-to-many).
- **Multiplicity:** Specifies the number of instances involved in a relationship (e.g., 1, 0.., 1..).

The diagram below illustrates the E-R Model for Therapy for treating phobias and Anxiety, showing the relationships between key entities and their attributes.

E-R Diagram

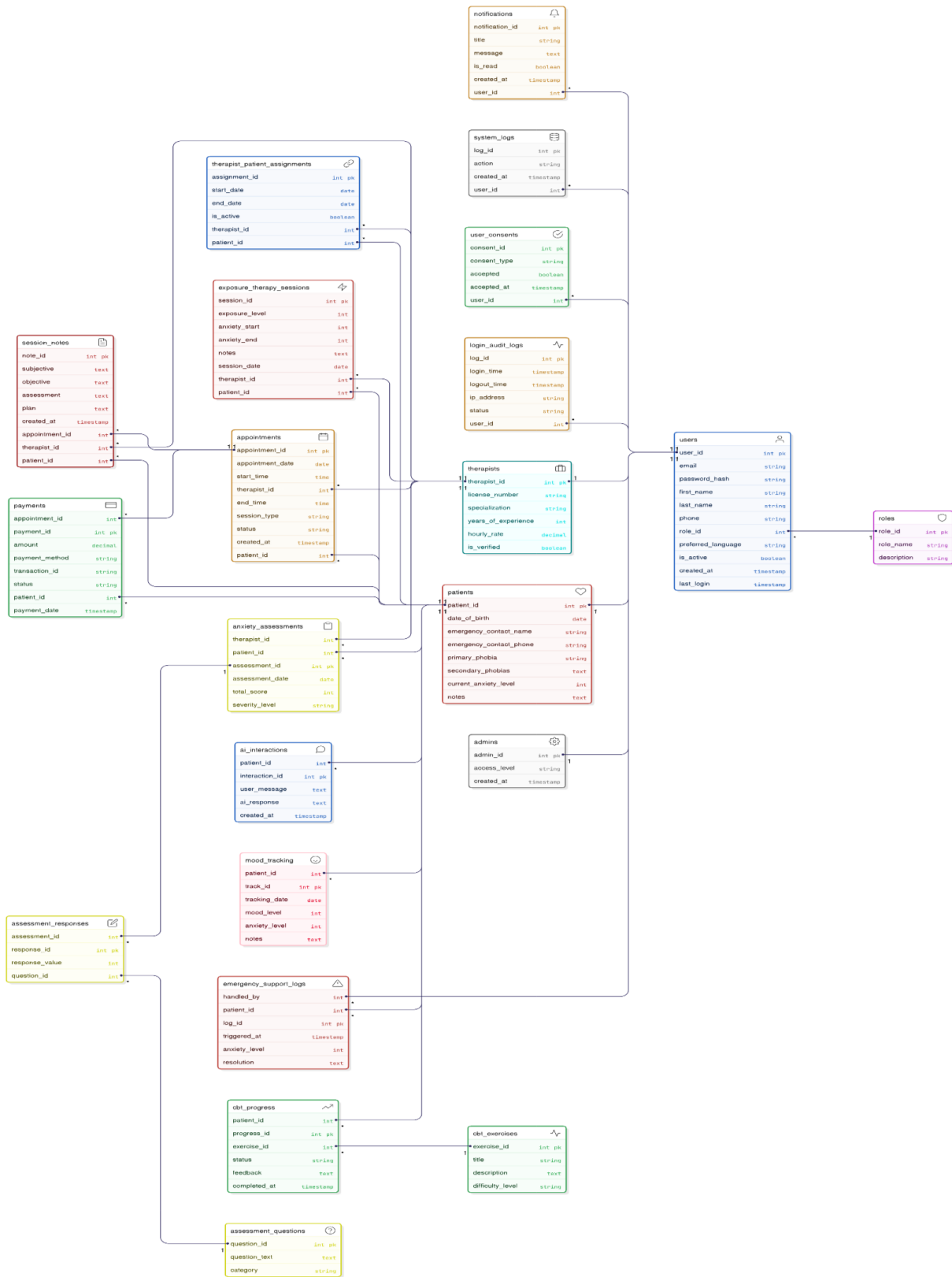


Figure 43: E-R Diagram

3.3 User Interface Design



Figure 44: Patient (Mobile app) UI

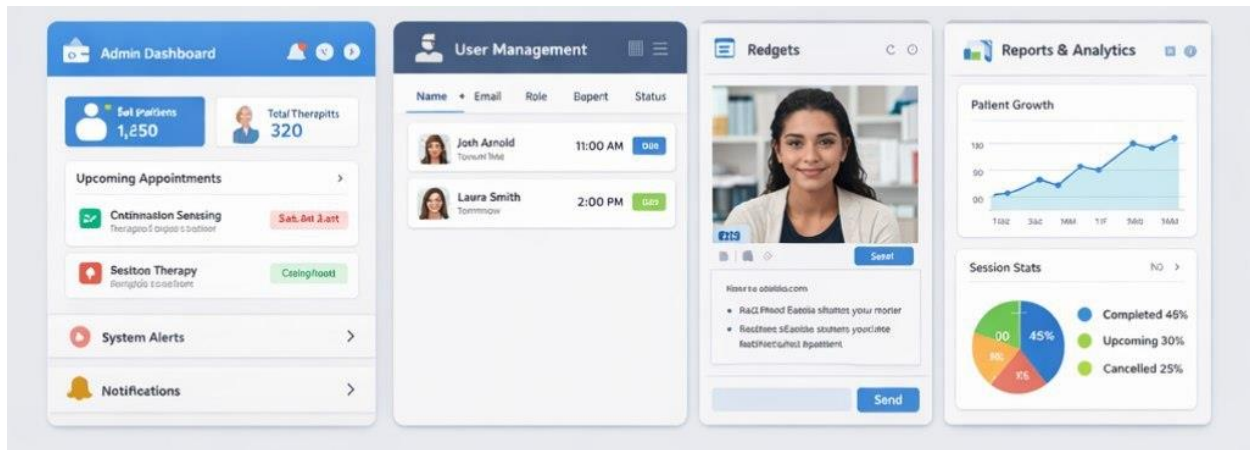


Figure 45: Admin UI

3.4 Access Control and Security

The Therapy Management System implements strong access control and security mechanisms to protect sensitive mental health information and ensure ethical handling of patient data. Since the system manages confidential psychological records and therapy sessions, strict security policies are enforced to prevent unauthorized access, data misuse, and privacy violations.

The system uses Role-Based Access Control (RBAC), where each user is assigned a specific role with clearly defined permissions. Access to system functions and data is granted strictly based on the user's role.

User Roles in the System

The system supports three main user roles:

1. Admin

- Manages the overall system.
- Controls user accounts and system configuration.
- Ensures compliance with security and ethical policies.

2. Therapist

- Provides therapy services to patients.
- Accesses only assigned patient data.
- Manages therapy sessions and clinical notes.

3. Patient

- Uses the system to receive therapy services.
- Accesses only personal data and therapy records.
- Interacts with therapists securely.

Access Control Matrix

System Functions / Tasks	Admin	Therapist	Patient
User registration & login	✓	✓	✓
Manage user accounts	✓		
Verify therapist credentials	✓		
View own profile	✓	✓	✓
Update own profile	✓	✓	✓
Access patient medical/therapy records		✓ (assigned)	✓ (own)
Conduct anxiety assessments			✓
View assessment results	✓	✓	✓
Schedule therapy sessions		✓	✓
Conduct therapy sessions		✓	
Record therapy notes		✓	
Access CBT exercises			✓
Access exposure therapy modules		✓	✓
Mood and progress tracking			✓

View progress reports	✓	✓	✓
Chat and secure communication		✓	✓
Emergency support handling	✓	✓	✓
System monitoring and maintenance	✓		

Table 18: Access control matrix

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